Sets to Real Analysis: A Thorough Reconstruction

Contents

Chapter 1

Introduction to Set Theory

1.1. Foundations and Framework

Logic

A **proposition** Q is anything that can be evaluated as true or false. If P, Q are propositions and P implies Q, we write $P \implies Q$. If P is implied by Q, we write $P \impliedby Q$. If P is true if and only if Q is true, we write $P \iff Q$. We write $\neg P$ to mean the negation of P, $P \wedge Q$ to mean P and Q, and $P \vee Q$ to mean P or Q (inclusively). The usual laws of logic will be assumed without explicating them further.

A **one-place predicate** P is anything that takes in an object a and outputs a proposition $P(a)$. For example, if $P(a)$ means, "a went to the store," then P refers to "went to the store." In this sense, P refers to something that is either true or false of any given object. In addition to predicates that take in a single object, we allow the following. A **two-place predicate** P takes objects a, b to form a proposition $P(a, b)$ and a **three-place predicate** P takes objects a, b, c to form a proposition $P(a, b, c)$. For example, if $P(a, b)$ means, "a is on top of b," then P refers to "is on top of."

When a discussion takes place, we delineate a **domain of discourse**. This refers to the class of everything that is under consideration in a discussion. For example, when we're talking about which animals make good pets, the domain of discourse might be limited to domesticated animals like cats, dogs, and rabbits instead of wild animals like tigers or whales. When we say, "all people have been interviewed," the domain of discourse might be job applicants for a specific position rather than all people in general. By writing $\forall x\, P(x)$, we mean: For all objects x, we have that $P(x)$. By writing $\exists x\, P(x)$, we mean: There exists an object x such that $P(x)$. Here $\forall$ is a **universal quantifier** and $\exists$ is an **existential quantifier**. In a statement like $\forall x\, P(x)$, the variable x is implicitly ranging over some domain like the set of natural numbers or a group of people.

Inside propositions with quantifiers, we use **variable symbols** like x, y, z, which do not refer to any specific object but are placeholders for arbitrary objects in a given domain of discourse. In general, a **bound variable** is a variable whose occurrence lies within the scope of something (a binding operator) that declares the variable role. The binding operators can be quantifiers, but later in the text we will come across integral, summation, and limit notations that also serve as binding operators. Bound variables act as local placeholders whose names may be changed without altering the meaning of the expression. A variable that is not bound is called a **free variable**.

In contrast to variables, we use **constant symbols** like a, b, c to refer to some specific objects in the domain of discourse, even if we might not know exactly which object. We will not always adhere to the convention of using initial English letters for constant symbols and final English letters for variable symbols; this will only be a heuristic and how any given letter is used should be understood from the context.

When we deal solely with propositions without predicates or quantifiers, we are dealing with **propositional logic**. When we deal with predicates and quantifiers that range over the objects of the domain of discourse, we are dealing with **first-order** or **predicate logic**. Most of this text will implicitly operate within informal predicate logic. The English language will be used throughout, with logical notation appearing at times only as a shorthand.

Equality

The most basic and prevalent relation in mathematics is the **equality relation** $=$. In mathematics, **terms** are expressions that designate objects in a given domain of discourse. If two terms, say a and b, refer to the same object, we write $a = b$. If the terms a and b refer to different objects, we write $a \neq b$ as a shorthand for $\neg(a = b)$. The following characterization of equality is sufficient for the mathematics we will be undertaking:

(1) Law of identity: For any object x in a given domain of discourse, $x = x$.
(2) Indiscernibility of identicals (schema): If P is a predicate, then for any objects x, y in a given domain of discourse, if $x = y$ then $P(x) \iff P(y)$.

The **indiscernibility of identicals** is also called the **substitution property**. In the way we formulated it, the indiscernibility of identicals is a schema rather than a single statement, so it represents multiple statements, one for each predicate P.

We will be flexible with our notation in obvious ways. For example, by writing $a = b = c$, we mean that $a = b$ and $b = c$ hold simultaneously. From properties (1) and (2) above, we may prove other properties of equality.

Proposition 1.1. *For any objects a, b, c in a given domain of discourse, we have the following:*

(a) Symmetry: *If $a = b$, then $b = a$.*
(b) Transitivity: *If $a = b$ and $b = c$, then $a = c$.*

Proof. Suppose $a = b$. Let $P(x)$ be the predicate $x = a$. By the law of identity, $P(a)$ holds true. By the substitution property and by the hypothesis that $a = b$, it follows $P(b)$ holds true, which stands for $b = a$.

Suppose $a = b$ and $b = c$. Let $Q(x)$ be the predicate $a = x$. By hypothesis, $Q(b)$ holds true. By the substitution property and by our assumption that $b = c$, it follows that $Q(c)$ holds true, which stands for $a = c$. ■

Oftentimes, we may have propositions of the form

$$\exists x \,(P(x) \land \forall y \,(P(y) \implies y = x))$$

for some predicate P. This asserts not only that an object a satisfies P, but also that the object is unique, so if b also satisfies P, then $a = b$. We may abbreviate this as

$$\exists! x \, P(x),$$

which is a statement of unique existence.

Constant, (Domain Guarded) Function, and Relation Symbols

In addition to connectives $\land$, $\lor$, $\implies$, $\impliedby$, $\iff$, quantifiers $\exists$, $\forall$, equality $=$, and parentheses, we will have three kinds of non-logical symbols.

A constant symbol c (already introduced above) is a symbol that refers to a fixed existing object in the domain of discourse. A **(domain guarded) function symbol** f with **domain condition** D is a symbol that takes in a term t for which a proof of $D(t)$ exists and returns a new term $f(t)$ that refers to an existing object in the domain of discourse. In some cases, the domain condition D may be a two-place predicate and the function symbol f may take two terms s, t for which a proof of $D(s, t)$ exists to produce a new term $f(s, t)$. A **binary relation symbol** R is a symbol that takes in two terms s, t and returns a proposition $R(s, t)$; we often write $R(s, t)$ as $s \, R \, t$.

The constant, function, and relation symbols we are discussing here are syntactic objects outside of the mathematical domain of discourse. If f is a function symbol with domain condition D, we only consider $f(t)$ well-formed if a proof of $D(t)$ exists (in the relevant context) where t is a well-formed term. A function f such that $f(t)$ is defined for all terms t is possible if the domain condition is trivial like $D(t) \equiv t = t$.

We deviate away from usual first-order logic because of our domain guarded scheme taken from Wiedijk and Zwanenburg's work. This is so that certain junk terms like $1/0$ (division by zero) remain genuinely undefined. Whenever we use a domain guarded function symbol, the relevant domain condition must already be available in the context. This has the consequence that propositions and predicates must be read from left to right such that the well-formedness of terms crucially depends on what is established to the left of the term and in the given context.

Example 1.2. Suppose our domain of discourse is the set of rational numbers. We haven't yet introduced rational numbers formally, but we use them here for illustrative purposes. Suppose division $d(x,y) = x/y$ is domain guarded by $D(x,y) \equiv y \neq 0$.

Consider the predicate
$$P(x) \equiv (x \neq 0) \wedge (1/x > 0).$$
If $x \neq 0$, the first clause is true and $D(1,x)$ holds, so then $1/x$ is well-formed and the second clause may be evaluated as true or false. If $x = 0$, the first clause is false, and we stop there immediately to conclude $P(x)$ is false without reading the second clause. Consider the predicate
$$Q(x) \equiv (1/x > 0) \wedge (x \neq 0).$$
Here, nothing ensures $x \neq 0$ when reading left to right, so no proof of $D(1,x)$ exists, and so $Q(x)$ is not well-formed.

The proposition
$$R \equiv \exists x\,(x \in \mathbb{Q} \wedge 1/x = 1)$$
is not well-formed because nothing ensures $x \neq 0$ to the left of $1/x$, so no proof of $D(1,x)$ exists. However,
$$S \equiv \exists x\,(x \in \mathbb{Q} \wedge x \neq 0 \wedge 1/x = 1)$$
is well-formed, because the $x \neq 0$ clause proves $D(1,x)$ before we get to $1/x$.

Axioms, Theory, and Models

Mathematics deals with structures such as numbers, shapes, graphs, spaces, and more complicated systems. These structures sit on the **semantic** side of mathematics. In order to reason about such structures, we write statements in a language. This language, together with the rules for forming statements and proofs done using logic, sits on the **syntactic** side of mathematics.

An **axiom** is a proposition we take as true within a given theory. Axioms in a given theory are used as the starting point so that we may prove further statements called theorems. Now if the axioms of a given theory are true about some mathematical structure(s), then by soundness of logic all the resulting theorems should also be true about the mathematical structure(s).

An **axiom schema** is a rule for generating axioms such that for each appropriate predicate P, the schema yields a unique proposition by having P substituted in the right places. It is simply a way to represent multiple axioms, one for each predicate.

Example 1.3. This example is adapted from Nagel and Newman. We take the formal language of two sorts K and L where variables x, y, z, w belong to K and variables i, j, k, l belong to L, and there is a single primitive relation symbol $\propto$. We read $x \propto i$ as "x is blorged in i." The theory consists of the following axioms:

(T1) Any two distinct K-objects are blorged in exactly one L-object:

$$\forall x \forall y \, (x \neq y \implies \exists! i \, (x \propto i \land y \propto i)).$$

(T2) No K-object is blorged in more than two L-objects:

$$\neg \exists x \exists i \exists j \exists k \, (i \neq j \land i \neq k \land j \neq k \land x \propto i \land x \propto j \land x \propto k).$$

(T3) The K-objects are not all blorged in a single L-object:

$$\neg \exists i \, \forall x \, (x \propto i).$$

(T4) Any two distinct L-objects blorg exactly one K-object:

$$\forall i \forall j \, (i \neq j \implies \exists! x \, (x \propto i \land x \propto j)).$$

(T5) No L-object blorgs more than two K-objects:

$$\neg \exists i \exists x \exists y \exists z \, (x \neq y \land x \neq z \land y \neq z \land x \propto i \land y \propto i \land z \propto i).$$

(T6) There exist two distinct K-objects:

$$\exists x \exists y \, (x \neq y).$$

Using these starting axioms of our theory, we can use logical inference to prove various theorems. For example, we may deduce that there must be exactly three K-objects, and all this can be done without having to interpret what any of the words or symbols mean.

However, the axioms we laid out happen to be true statements about a real mathematical object. In our example, we characterized a triangle-like structure. We may interpret K-objects as vertices, L-objects as line segments or edges, and the blorg relation $\propto$ to mean that a vertex is part of the line segment or edge. Since (T1) to (T6) are true of triangles, any deductions from them such as the theorem that there are exactly three K-objects are also true of triangles.

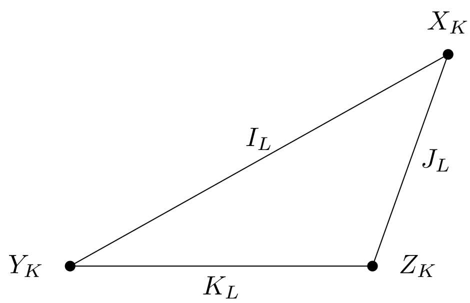

Figure 1.1: Triangle model of the axioms.

Axiom lists may characterize a class of structures. Sometimes they may narrow the class down to exactly one structure, but this does not have to be true. For example, if we removed some axioms from Example 1.3, then the triangle would not be the only compatible structure satisfying the axioms.

Extending Our Language and Theory

In our text, we will be extending our language and theory as our text unfolds exactly as in real life mathematical practice. Let us explain what this means. Let L be a **language** containing logical symbols along with constant, function, and relation symbols, and let T be a **theory-as-a-set-of-axioms** in L. We say L' **extends** L if every symbol of L is also a symbol of L'. We say T' **extends** T if every axiom of T is also an axiom of T'.

In this text, extensions arise by adding new constant, function, or relation symbols, together with axioms governing them. However, we require that every extension T' in an expanded language L' is **conservative** over T, meaning that every proposition expressible in the original language L that is provable from T' must be provable from the original theory T. This way, what is and is not entailed by our original axioms is preserved. We do not want to add symbols and axioms that break conservativity, because then the extension would leave us with a theory that is inequivalent to the original theory. In turn that would change what mathematical structures we are talking about.

Consider the following scenario. If our initial theory proves $\neg\exists x\, P(x)$, then adding a new constant symbol c such that $P(c)$ holds introduces a contradiction, because the existence of c implies $\exists x\, P(x)$. This breaks not only conservativity over the original theory but consistency itself. In this case, the expansion of language and the introduction of new notation is not harmless. Let us lay out the precise cases where expansion of language and theory is conservative.

- Introducing constant symbols. Constants refer to existing mathematical objects. If we show
$$\exists! x\, P(x)$$
for some predicate P, then we may introduce a new symbol c with a new axiom
$$P(c). \tag{1.1}$$
Since we require uniqueness, if d satisfies P, it follows that $c = d$.
- Introducing (domain guarded) function symbols. Function symbols are used to transform terms into other terms. If we show
$$\forall x\, (D(x) \implies \exists! y\, P(x, y))$$
for some one-place predicate D and two-place predicate P, then given x such that $D(x)$ holds we want to be able to pick out y such that $P(x, y)$ holds. We do this by introducing a new symbol f with domain condition D and a new axiom
$$\forall x\, (D(x) \implies P(x, f(x))). \tag{1.2}$$
Since uniqueness of outputs must be established, if c, d are such that $D(c)$ and $P(c, d)$, it must follow that $d = f(c)$. Sometimes we will also have D as a two-place predicate and P as a three-place predicate so that given x, y for which $D(x, y)$ holds, we may form $f(x, y)$ so that $P(x, y, f(x, y))$ holds.

- Introducing binary relation symbols. Binary relation symbols take two terms and return a proposition. In general, for any two-place predicate P, we may add a binary relation symbol R with a new axiom

$$\forall x \forall y\,(P(x,y) \iff xRy). \tag{1.3}$$

All three cases are **extensions by definition**. In an extension by definition, any individual formula in the expanded language L' can be translated into a provably equivalent formula in the old language L. The translation is done on the atomic formulas and terms containing the new symbols. For example, if $\exists! x\, P(x)$ holds and c is a constant symbol such that (1.1) holds, then in any formula we may replace every atomic formula of the form $Q(c)$ with $\exists x\,(P(x) \wedge Q(x))$ to get an equivalent formula. If a function symbol f is introduced under (1.2) with uniqueness of outputs, then terms involving f are eliminated from the innermost outward: At each step, we rewrite each atomic formula $Q(f(t))$ with t not containing f as $\exists y\,(P(t,y) \wedge Q(y))$, provided $D(t)$ holds. By repeating this and working outwards, every occurrence of f is removed. If a relation R is introduced by (1.3), then every occurrence of sRt may be replaced by $P(s,t)$ where s,t are terms. Consequently, any proof involving c, f, or R introduced by extension by definition may be rewritten without those symbols. This guarantees the extension T' of T is conservative.

There are other ways in which constant symbols are introduced, but they are only done in *local contexts* (like in a proof or a theorem statement) and the *global language* does not change. Let us go over ways this happens in proofs.

- Universal generalization. Suppose we want to prove

  $$\forall x\, P(x)$$

  for some predicate P. To prove this, we would introduce a constant symbol c with nothing assumed other than background information like the axioms of our theory. After that, we use the axioms and logical inference to deduce $P(c)$. Because the object c refers to is arbitrary, we generalize, and infer $\forall x\, P(x)$, with c no longer part of our vocabulary. As an instructive example, suppose $P(x)$ is an implication of the form $R(x) \implies S(x)$. We introduce c and consider two cases. If c does not satisfy R, then $R(c) \implies S(c)$ is true by virtue of the antecedent being false. If c satisfies R, then we use axioms and logical inference to deduce that c satisfies S, so that $R(c) \implies S(c)$ is true by virtue of the consequent being true. Since this applies to all c, we generalize and conclude $\forall x\,(R(x) \implies S(x))$, with c no longer part of our vocabulary.
- Existential instantiation. Suppose we want to prove some proposition Q using a previously established fact that $\exists x\, P(x)$ where P is some predicate. To proceed, we introduce a constant symbol c such that $P(c)$, do reasoning in the given context, reach conclusion Q, and then discharge c. The constant symbol c must not appear anywhere before its introduction or in Q. It is important to stress that c is a *generic witness* to the existential statement $\exists x\, P(x)$. This means the only fact we may use

about c is that it satisfies P and anything entailed by that, even though a *specific object* might satisfy many more predicates.

In these local contexts, we do not need uniqueness conditions. For example, introducing a witness c such that $P(c)$ can be done if $\exists x\, P(x)$ is established with uniqueness not necessarily established (so if d is also such that $P(d)$ it does not necessarily follow that $c = d$). Proofs that invoke temporary constant symbols are translatable into proofs that avoid this technique. However, individual sentences with c cannot be translated into equivalent sentences without c unless the uniqueness condition $\exists! x\, P(x)$ also holds. (Proving all these facts is outside the scope of this text.) This is why we limit changes of our global language to only cases where uniqueness conditions hold.

There are formal proof systems that make all this much more precise. In some systems like natural deduction, local introductions of new constant symbols are part of the proof-theoretic framework, but there are systems like Hilbert's system that avoid any introduction of constant symbols altogether. The price to pay in systems like Hilbert's system is that proofs are much more arduous and difficult to follow.

In our text, we will sometimes invoke another layer to our discussion, which involves talking about the syntax explicitly. A **character** is taken to be any symbol in a language. More generally, it can be any atomic piece of a writing system that can't be broken down further. A **string** is any sequence of characters written together consecutively. We include the **empty string**, which consists of no characters, as a string. In our framework, strings and characters will be linguistic devices sitting outside of our object-level mathematics.

Just like how strings denote mathematical objects, we may sometimes introduce meta-variables and meta-constants to denote strings. For example, we may let $\underline{S}$ denote the string 00517, which is a numeral distinct from the number 00517. Equality $=$ is reserved for object-level terms, while $\equiv$ will be a meta-relation used to denote equality of propositions, predicates, and strings. So going to our numeral vs number example, we have $00517 = 517$ but $00517 \not\equiv 517$, because the latter is a statement about uninterpreted strings.

1.2. Basic Set Axioms

The Conception of Set Theory

Set theory provides powerful tools to organize mathematical concepts. Additionally, set theory is capable of subsuming and encoding a large amount of mathematics, as we will see in our text. The theory we will be working with shall be **Zermelo–Fraenkel set theory with the axiom of choice**, which is abbreviated as **ZFC**. The domain of ZFC consists of only one sort, which is the **set**, so all elements of ZFC are sets which themselves contain other sets and so on.

Intuitively, the universe of sets may be viewed as being generated in stages, beginning with the empty set $\varnothing$. Since $\varnothing$ exists, the set $\{\varnothing\}$ containing only $\varnothing$ may be formed; given these we may form sets such as $\{\{\varnothing\}\}$ and $\{\varnothing, \{\varnothing\}\}$, and so on. Continuing this process through both finite and infinite stages yields the cumulative hierarchy of sets. All steps are specified by the axioms of ZFC.

We will operate with two layers. One layer will involve specifying the structure of mathematical concepts and the other layer will involve implementing the concept within set theory. For example, we might define an ordered pair (a, b) to be an object o whose only relevant information is the objects a and b and the order in which they occur. This is sufficient as a definition in our first layer, but because everything is a set in the second layer, we will provide a set-theoretic realization of this type of object; an example we will later see is the Kuratowski implementation in which $o = \{\{a\}, \{a, b\}\}$. We may specify a theory of a number system axiomatically and then work with that given theory, but in addition to this we will show how the described number system is realized within pure set theory (we will provide a set-theoretic **model** for the given **theory**). The motivation for the latter part is a matter of demonstrating relative consistency. For many theories (such as Peano arithmetic), we cannot establish their consistency in an absolute sense, but we can establish relative consistency by providing set-theoretic realizations of what the theories are describing. If we are reasonably confident that ZFC is consistent, then an implementation of some number system within set theory means we can be reasonably confident the given theory is consistent as well.

Our philosophy will be along the lines of **structuralism**, which is the view that mathematical objects are defined by the roles they play within a structure, rather than by some intrinsic nature they possess independently. Although every mathematical object discussed in this text will be realized as a set, our interest will be in the properties and relationships of mathematical objects rather than in what those objects are. A strong case for structuralism can be made by pointing out that a given concept can be realized in set theory in more than one way and that there are alternative competing foundations of mathematics that are distinct from set theory as well.

Sets, Membership, and Subset Relations

In ZFC, any mention of element or object is taken as synonymous with set. The base formal language of ZFC consists of the logical symbols and one additional relation symbol $\in$. However, as we explained above, we will not adhere to this language and we will introduce constant terms, (domain guarded) function symbols, and relation symbols as we see fit. The $\in$ relation represents **membership**, so if a, b are sets and $a \in b$ then we say a is a **member** or **element** of b. If it is not the case that $a \in b$ then we write $a \notin b$.

Let us now provide two axioms.

(ZFC1) Axiom of the Empty Set. There exists a set $\varnothing$ such that $\forall x\,(x \notin \varnothing)$.

(ZFC2) Axiom of Extensionality. For any A, B, if $\forall x\,(x \in A \iff x \in B)$, then $A = B$.

The axiom of the empty set asserts the existence of an **empty set**, defined as any set $\varnothing$ satisfying $\forall x\,(x \notin \varnothing)$.

Proposition 1.4. *The empty set is unique.*

Proof. Suppose $\varnothing, \varnothing'$ have the property that $x \notin \varnothing$ and $x \notin \varnothing'$ for all x. Then for all x, the equivalence $x \in \varnothing \iff x \in \varnothing'$ is vacuously true, so $\varnothing = \varnothing'$ by (ZFC2). ■

We will universally denote the empty set by $\varnothing$ or $\{\}$. Since it is unique, there is no ambiguity as to what these symbols refer to.

If $A = B$, then by the substitution property it is necessarily the case that $\forall x\,(x \in A \iff x \in B)$, which means A and B have exactly the same elements. However, the converse does not necessarily follow without the axiom of extensionality. The axiom of extensionality asserts that sets are defined by nothing other than what they contain, so if A and B contain exactly the same elements, then $A = B$. (We could imagine perhaps in an alternative version of set theory, sets come in different colors, so A and B would have the same elements but A would be a red set and B would be a green set, thus rendering $A \neq B$. The axiom of extensionality ensures this does not happen.)

Definition. Suppose A and B are sets. If every element of A is an element of B, we say A is a **subset** of B and write $A \subseteq B$. Equivalently, we say B is a **superset** of A and write $B \supseteq A$. If A is a subset of B but not equal to B, we say A is a **proper subset** of B and write $A \subset B$. Equivalently, we say B is a **proper superset** of A and write $B \supset A$.

Proposition 1.5. *For any set A, we have $\varnothing \subseteq A$ and $A \subseteq A$.*

Proof. For any x, the implication $x \in \varnothing \implies x \in A$ is vacuously true, so $\varnothing \subseteq A$. For any x, the implication $x \in A \implies x \in A$ is tautologically true, so $A \subseteq A$. ■

Suppose A and B are sets. If $A = B$, then by the substitution property of equality, $x \in A \iff x \in B$ for all x, so the subset relations $A \subseteq B$ and $B \subseteq A$ both hold. We leave it to the reader to show the converse: If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

We will often abbreviate certain logical sentences in obvious ways. For example, given a set A, instead of writing

$$\forall x\,(x \in A \implies P(x)) \quad \text{and} \quad \exists x\,(x \in A \land P(x)),$$

we write

$$\forall x \in A : P(x) \quad \text{and} \quad \exists x \in A : P(x)$$

and variants thereof.

Construction Axioms

The following axioms describe how to build new sets out of existing objects. We will use these axioms to define various set operations.

(ZFC3) Axiom of Pairing. Given sets a, b, there exists a set

$$B = \{a, b\}$$

for which $x \in B$ if and only if $x = a$ or $x = b$. Intuitively, B is the set consisting of elements a and b only, and we call this the **unordered pairing** of a and b.

(ZFC4) Axiom of Union. Given a set A, there exists a set

$$B = \bigcup A$$

such that $x \in B$ if and only if $x \in y$ for some $y \in A$. Intuitively, B is the set obtained by taking all members of A and then putting the elements of those members into a single set, and we call this the **union** of elements of A.

(ZFC5) Axiom schema of separation. Given sets A, c and a (one-place or two-place) predicate P, there exists a set

$$B = \{x \in A \mid P(x, c)\}$$

such that $x \in B$ if and only if $x \in A$ and $P(x, c)$. Intuitively, B is the subset of A consisting of elements that have the property P (with parameter c), and we call this the **specified subset** of A of elements satisfying predicate P (with parameter c).

(ZFC6) Axiom of Power Set. Given a set A, there exists a set

$$B = \mathcal{P}(A)$$

such that for all X we have $X \subseteq A \iff X \in B$. Intuitively, B is the set of all subsets of A, and we call this the **power set** of A.

The axiom schema of separation also goes by the names **specification**, **subsets**, and **restricted comprehension**.

The axiom schema of separation depends on predicates defined in the original language consisting only of the non-logical $\in$ symbol. However, because all expansions of our language will be extensions by definition (as explained in Section 1.1), we may invoke predicates that use the expanded language without altering the theory whatsoever.

The construction axioms assert existence of various sets. Let us go over some basic uniqueness propositions.

Proposition 1.6. *We have the following:*

(a) *Given sets a, b, the set B asserted by* (ZFC3) *is unique.*
(b) *Given a set A, the set B asserted by* (ZFC4) *is unique.*
(c) *Given sets A, c and a predicate P, the set B asserted by* (ZFC5) *is unique.*
(d) *Given a set A, the set B asserted by* (ZFC6) *is unique.*

Proof. Given sets a, b, suppose B, B' both satisfy the description given by (ZFC3). Then $x \in B$ if and only if $x = a \vee x = b$ if and only if $x = b \vee x = a$ if and only if $x \in B'$. By the axiom of extensionality (ZFC2), $B = B'$.

Given a set A, suppose B, B' both satisfy the description given by (ZFC4). Then $x \in B$ if and only if $\exists y\,(y \in A \wedge x \in y)$ if and only if $x \in B'$. By the axiom of extensionality (ZFC2), $B = B'$.

Given sets A, c and a predicate P, suppose B, B' both satisfy the description given by (ZFC5). Then $x \in B$ if and only if $x \in A \wedge P(x, c)$ if and only if $x \in B'$. By the axiom of extensionality (ZFC2), $B = B'$.

Given a set A, suppose B, B' both satisfy the description given by (ZFC6). Then $X \in B$ if and only if $X \subseteq A$ if and only if $X \in B'$. By the axiom of extensionality (ZFC2), $B = B'$. ∎

Now that uniqueness is established, we may use the notation invoked in (ZFC3) to (ZFC6) universally. Given that

$$\forall a \forall b \exists! B\, Q(a, b, B) \quad \text{where} \quad Q(a, b, B) \equiv \forall x\,(x \in B \iff x = a \vee x = b),$$

we introduce a function symbol Pair such that $\mathrm{Pair}(a, b)$ is B in (ZFC3). After that, we rewrite $\mathrm{Pair}(a, b)$ as $\{a, b\}$ for readability. Given a predicate P, we have

$$\forall A \forall c \exists! B\, Q(A, c, B) \quad \text{where} \quad Q(A, c, B) \equiv \forall x\,(x \in B \iff x \in A \wedge P(x, c)),$$

so we may introduce a function symbol Sep_P such that $\mathrm{Sep}_P(A, c)$ is B in (ZFC5). After that, we rewrite $\mathrm{Sep}_P(A, c)$ in **set-builder notation**: $\mathrm{Sep}_P(A, c) = \{x \in A \mid P(x, c)\}$. For (ZFC4) and (ZFC6), we introduce function symbols $\bigcup$ and $\mathcal{P}$ accordingly.

Let us give an interesting application of the axioms we have so far. This famous result concerns the non-existence of a certain set in ZFC set theory.

Proposition 1.7 (Russell's paradox)**.** *The set of all sets does not exist.*

Proof. We proceed by proof by contradiction. Suppose such a set A exists. Let $P(x) \equiv x \notin x$ and apply separation to get the subset $B = \{x \in A \mid x \notin x\}$. If $B \in B$, then the definition of B forces $B \notin B$. If $B \notin B$, then the definition of B forces $B \in B$. In all cases, we obtain a contradiction. Since the existence of B is entailed by the existence of A, our supposition that A exists must be false. ∎

Using the construction axioms we introduced, we may start to define new kinds of set operations.

Definition (Binary union)**.** Let A and B be sets. We denote the **binary union of** A **and** B by

$$A \cup B = \bigcup \{A, B\}.$$

Definition (Binary intersection)**.** We define the **binary intersection of** A **and** B to be the set

$$A \cap B = \{x \in A \mid x \in B\}.$$

Definition (Set difference)**.** We define the **set difference of** A **relative to** B, also referred to as the **relative complement of** B **in** A, to be the set

$$A \setminus B = \{x \in A \,|\, x \notin B\}.$$

Let us use the axioms of set theory to establish that all three set operations form valid sets out of existing sets.

Proposition 1.8. *For any sets A and B, the sets $A \cup B$, $A \cap B$, and $A \setminus B$ exist and are unique.*

Proof. By the axiom of pairing (ZFC3), there exists the set $\{A, B\}$. By the axiom of union (ZFC4), there exists a set U such that $x \in U$ if and only if $x \in Y$ for some $Y \in \{A, B\}$. The condition $Y \in \{A, B\}$ is precisely equivalent to $Y = A$ or $Y = B$, so then the statement $x \in Y$ is simply $x \in A$ or $x \in B$. This shows U is precisely the set consisting of all elements x such that $x \in A$ or $x \in B$. Thus, U is a binary union of A and B, so $A \cup B$ exists.

To obtain the intersection $A \cap B$, apply the axiom schema of separation (ZFC5) to A with the predicate $P(x) \equiv x \in B$. To obtain the set difference $A \setminus B$, apply the axiom schema of separation (ZFC5) to A with the predicate $Q(x) \equiv x \notin B$.

Uniqueness of all three sets given A and B is left to the reader to prove. ■

If $A \cap B = \varnothing$, we say A and B are **disjoint** or equivalently they have an **empty intersection**. Otherwise, we say A and B **intersect** or equivalently they have a **nonempty intersection**.

Given objects a, b, the order in which we write the elements in $\{a, b\}$ does not matter, because one can show $\{a, b\} = \{b, a\}$; it is the set containing exactly the objects a and b in no particular order. To see this, simply note that $x \in \{a, b\}$ if and only if $x = a \lor x = b$ if and only if $x = b \lor x = a$ if and only if $x \in \{b, a\}$.

Given an element a, we may use the axiom of pairing (ZFC3) to form the set $\{a, a\}$. Since $x \in \{a, a\}$ if and only if $x = a \lor x = a$ if and only if $x = a$, this set contains only the element a. If A is a set such that $\forall x\, (x \in A \iff x = a)$, we call A a **singleton**, and the axiom of extensionality (ZFC2) implies $A = \{a, a\}$, which is unique with respect to a. For any a, the singleton containing a is written as $\{a\}$.

Given two singletons $\{a\}$ and $\{b\}$, we may form the union $\{a\} \cup \{b\}$ and get back the pairing $\{a, b\}$ of a and b. We may write $\{a, b\} \cup \{c\}$ as $\{a, b, c\}$. A string $\underline{S}$ lists out all elements of a set if it is either of the form $\{\underline{a}\}$ or $\underline{T}, \underline{a}\}$ where $\underline{a}$ is a string denoting an element and $\underline{T}\}$ lists out all elements of another set. If $\underline{S}\}$ lists out all elements of a set, then the set denoted by the string $\underline{S}\} \cup \{\underline{a}\}$ may be also denoted by the string $\underline{S}, \underline{a}\}$, which lists out all elements of the set. For example, if

$$\underline{S} \equiv \{a, b, c$$

then

$$\underline{S}\} \equiv \{a, b, c\}$$

and then $\{a, b, c\} \cup \{d\}$ is rewritten as $\{a, b, c, d\}$.

Example 1.9 (Union)**.** Suppose

$$A = \{\{a, b, c\}, \{b, c, d\}, \{a, e\}\}.$$

Then the union of elements of A is

$$\bigcup A = \{a, b, c, d, e\}.$$

Example 1.10 (Separation)**.** Suppose i, j, k are distinct elements and

$$A = \{a, b, c, d, e\} \quad \text{where} \quad a = \{i, j\},\ b = \{i\},\ c = \{j\},\ d = \{i, j, k\},\ e = \{j, k\}.$$

Let $P(x) \equiv i \in x$ and

$$Q(x) \equiv \exists y \exists z\, (y \neq z \wedge y \in x \wedge z \in x \wedge \forall w\, (w \in x \implies (w = y \vee w = z))),$$

so $P(x)$ says x contains element i and $Q(x)$ says x contains exactly two elements. By applying separation, we obtain sets

$$\{x \in A \mid P(x)\} = \{a, b, d\} \quad \text{and} \quad \{x \in A \mid Q(x)\} = \{a, e\}.$$

Example 1.11 (Power set)**.** Let $A = \{a, b, c\}$ with a, b, c distinct from one another. Then the power set of A is

$$\mathcal{P}(A) = \{\{a, b, c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a\}, \{b\}, \{c\}, \varnothing\}.$$

Let us provide some examples showcasing binary set operations.

Example 1.12. Let $A = \{a, b, c\}$, $B = \{j, k\}$, and $C = \{b, c, k\}$ where a, b, c, j, k are distinct elements. Then

$$\begin{aligned} A \cup B &= \{a, b, c, j, k\}, & A \cup C &= \{a, b, c, k\}, \\ A \cap B &= \{\}, & A \cap C &= \{b, c\}, \\ A \setminus B &= \{a, b, c\}, & A \setminus C &= \{a\}. \end{aligned}$$

The binary union operation can be considered as a special case of the general union operator from the axiom of union (ZFC4). In a similar manner, the binary intersection operation can be considered as a special case of a more general intersection operator defined below.

Definition (Intersections)**.** Given a nonempty set A, we define the **intersection** of elements of A as

$$\bigcap A = \{x \in \bigcup A \mid \forall y\, (y \in A \implies x \in y)\}.$$

This is the set of elements x such that $x \in y$ for all $y \in A$.

If A is nonempty, the existence of $\bigcap A$ follows from the axiom of union (ZFC4) and the axiom schema of separation (ZFC5). If A is empty, we leave $\bigcap A$ undefined, so $\bigcap$ is a domain guarded function symbol with domain condition $D(x) \equiv x \neq \varnothing$. (There is nothing wrong with applying union followed by separation in the case where A is empty, but it is conventional to leave the intersection undefined. The thinking is that we want to preserve the property that $A \subseteq B$ implies $\bigcap A \supseteq \bigcap B$ for all applicable sets A, B. If $\bigcap \varnothing$ were defined such that the subset property held, then the fact that $\varnothing \subseteq A$ for all sets A would imply $\bigcap \varnothing$ contains all sets. However, such a set does not exist by Russell's paradox.)

Kuratowski–Cartesian Products

Recall that the order in which we write the elements in the pair $\{a, b\}$ does not matter. Before we define a particular type of set-theoretic construction of an **ordered pair**, let us give a characterization of what we want. Given any two elements a, b we want to be able to form an object $o = \mathrm{OrdPr}(a, b)$ with two operations π_L and π_R that recover the original elements: $\pi_L(o) = a$ and $\pi_R(o) = b$. We also want that if o, o' are two such objects such that $\pi_L(o) = \pi_L(o')$ and $\pi_R(o) = \pi_R(o')$, then $o = o'$. This way, $o = \mathrm{OrdPr}(a, b)$ with π_L and π_R is the object whose information content is the first element a and the second element b.

Let us proceed with a set-theoretic construction. Given sets a, b, repeated applications of the axiom of pairing (ZFC3) allow us to form the set $\{\{a\}, \{a, b\}\}$. We call this a **Kuratowski ordered pair** and denote it as $\langle a, b\rangle$. We can get back elements a and b from $\langle a, b\rangle$ as follows. For the first element, we find

$$\bigcup\bigcap\langle a, b\rangle = \bigcup\bigcap\{\{a\}, \{a, b\}\} = \bigcup\{a\} = a.$$

For the second element, we claim

$$\bigcup\left\{x \in \bigcup\langle a, b\rangle \,|\, P(x, \langle a, b\rangle)\right\} = b \quad \text{where} \quad P(x, c) \equiv \bigcup c \neq \bigcap c \implies x \notin \bigcap c.$$

To see this, note that if $a = b$, then $\bigcup\langle a, b\rangle = \bigcap\langle a, b\rangle$ so $P(x, \langle a, b\rangle)$ is true for all x, and the set in the set-builder notation is $\{x \in \{b\} \,|\, P(x, \langle a, b\rangle)\} = \{b\}$, and thus the left-hand side evaluates to b. If $a \neq b$, then $\bigcup\langle a, b\rangle \neq \bigcap\langle a, b\rangle$ so $P(x, \langle a, b\rangle)$ is true only for $x \notin \bigcap\langle a, b\rangle = \{a\}$, i.e. $x = b$, so the set in the set-builder notation is $\{x \in \{a, b\} \,|\, x = b\} = \{b\}$ and thus the left-hand side evaluates to b. Consequently,

$$\langle a, b\rangle = \langle a', b'\rangle \iff a = a' \text{ and } b = b'. \tag{1.4}$$

If $a \neq b$ then $\langle a, b\rangle \neq \langle b, a\rangle$, so the order in which we write a and b here matters. For any pair $o = \langle a, b\rangle$, we'll write $\pi_L(o) = a$ and $\pi_R(o) = b$ according to the above operations used.

Definition. Given sets A, B, we define the **binary Kuratowski–Cartesian product** to be the set

$$A \times_K B = \{z \in \mathcal{P}(\mathcal{P}(A \cup B)) \,|\, z = \langle x, y\rangle \text{ for some } x \in A \text{ and } y \in B\}.$$

In other words, $A \times_K B$ is the set of all Kuratowski ordered pairs $\langle x, y\rangle$ where $x \in A$ and $y \in B$.

The existence of $A \times_K B$ given sets A, B follows straightforwardly from the existence of $A \cup B$, the axiom of power set (ZFC6), and the axiom schema of separation (ZFC5). If A, B are nonempty, there exist elements $a \in A$ and $b \in B$, so $\langle a, b \rangle \in A \times_K B$ and $A \times_K B$ is nonempty.

Replacement and Foundation Axioms

The following axioms are meant to further specify what kinds of sets we are allowed to have and what kinds of sets we are not allowed to have.

(ZFC7) Axiom Schema of Replacement. Given a three-place predicate P such that

$$\forall x \forall c \exists! y \, P(x, y, c), \tag{1.5}$$

and sets A, c, there exists a set B such that

$$\forall x \, (x \in A \implies \exists y \, (y \in B \wedge P(x, y, c))) \quad \text{and} \quad \forall y \, (y \in B \implies \exists x \, (x \in A \wedge P(x, y, c))).$$

(ZFC8) Axiom of Foundation/Regularity. If A is a nonempty set, there exists a set $x \in A$ such that $x \cap A = \varnothing$.

Any predicate P satisfying (1.5) is called a **functional predicate**.

Proposition 1.13. *Given sets A, c and a functional predicate P, the set B asserted by* (ZFC7) *is unique.*

Proof. Given sets A, c and functional predicate P, suppose B, B' both satisfy the description given by (ZFC7). Let b be any element of B, so there exists $a \in A$ such that $P(a, b, c)$. Then there exists an element $b' \in B'$ such that $P(a, b', c)$. Since P is a functional predicate, $b = b'$ and so $b \in B'$. This shows $B \subseteq B'$. By the same reasoning, $B' \subseteq B$, so by the axiom of extensionality (ZFC2), $B = B'$. ∎

Just like with the axiom schema of separation, the axiom schema of replacement depends on predicates defined in the original language consisting only of the non-logical $\in$ symbol, but since all expansions of our language are extensions by definition, we may invoke predicates that use the expanded language without altering the theory whatsoever.

Given a functional predicate P and sets A, c, the existence and uniqueness of the set given by replacement may be written as $B = \mathrm{Rep}_P(A, c)$, or in set-builder notation,

$$B = \{\mathcal{F}_P(x, c) \mid x \in A\}$$

where $\mathcal{F}_P$ is the function corresponding to the functional predicate P.

Proposition 1.14. *For all sets A, $A \notin A$.*

Proof. Let A be any set. By the axiom of pairing (ZFC3), we may form the set $\{A\}$. Then the axiom of foundation (ZFC8) implies there exists $x \in \{A\}$ for which $x \cap \{A\} = \varnothing$. In this case, $x = A$ so $A \cap \{A\} = \varnothing$. If $A \in A$, then $A \in A \cap \{A\}$, which contradicts the previous sentence. Thus, $A \notin A$. ∎

1.3. Set-Coded Binary Relations

Basic Definitions

We introduced equality $=$ as a logical relation and membership $\in$ as a primitive relation. We also introduced subset relations $\subseteq, \supseteq, \subset, \supset$, which act as shorthands for predicates that we could in principle write out without those subset symbols. Now we want to introduce other kinds of relations, but we want to treat them as objects of set theory itself. For this reason, we will introduce new relations by encoding them as sets.

Definition. A **(set-coded) binary relation** R from A to B is a set of the form

$$R = \langle G_R, \langle A, B \rangle \rangle \quad \text{where} \quad G_R \subseteq A \times_K B.$$

If $\langle a, b \rangle \in G_R$, then we say a **is related to** b **by relation** R, and we write $a\,R\,b$. If $\langle a, b \rangle \notin G_R$, then we say a **is not related to** b **by relation** R, and we write $a \not{R}\, b$.

If R is a relation from A to itself, we say R is a relation on A. Most relations we will come across will be on one set. Note that it matters which element appears first and which appears second in the ordered pair $\langle a, b \rangle$, even in the case $B = A$, so we might have $a\,R\,b$ even though $b \not{R}\, a$.

Example 1.15. Let $A = \{a, b, c, d\}$ where a, b, c, d are distinct elements. Let P be the predicate

$$P(x) \equiv x = \langle a, b \rangle \vee x = \langle b, c \rangle \vee x = \langle c, d \rangle \vee x = \langle b, a \rangle \vee x = \langle c, b \rangle.$$

By the existence of $A \times_K A$ and separation, we obtain the set $G_\sim = \{x \in A \times_K A \mid P(x)\}$, and we form $\sim = \langle G_\sim, \langle A, A \rangle \rangle$. Since $G_\sim \subseteq A \times_K A$, this is a relation. We may list all the relationships by $\sim$ as follows:

$a \not\sim a$	$a \sim b$	$a \not\sim c$	$a \not\sim d$
$b \sim a$	$b \not\sim b$	$b \sim c$	$b \not\sim d$
$c \not\sim a$	$c \sim b$	$c \not\sim c$	$c \sim d$
$d \not\sim a$	$d \not\sim b$	$d \not\sim c$	$d \not\sim d$

Total Order Relations

Here we introduce two closely related types of relations.

Definition. A **strict total order relation** on A is a relation $<$ with the following properties for all $a, b, c \in A$:

(T1) Irreflexivity: $a \not< a$,

(T2) ASYMMETRY: if $a < b$ then $b \not< a$,
(T3) TRANSITIVITY: if $a < b$ and $b < c$, then $a < c$,
(T4) TRICHOTOMY: $a < b$, $a = b$, or $b < a$.

A **weak total order relation** on A is a relation $\leq$ with the following properties for all $a, b, c \in A$:

(T1′) REFLEXIVITY: $a \leq a$,
(T2′) ANTISYMMETRY: if $a \leq b$ and $b \leq a$, then $a = b$,
(T3′) TRANSITIVITY: if $a \leq b$ and $b \leq c$, then $a \leq c$,
(T4′) COMPARABILITY: $a \leq b$ or $b \leq a$.

Given $<$ we may define a corresponding $\leq$ and vice-versa by declaring $\forall x \forall y\,(x \leq y \iff x < y \vee x = y)$ and $\forall x \forall y\,(x < y \iff x \leq y \wedge x \neq y)$. We also refer to total order relations as **linear order relations**.

If A has a strict or weak total order relation, then we say A is a **totally ordered set** or **linearly ordered set** and introduce it as a pair $\langle A, < \rangle$. If A has a strict $<$ and a corresponding weak $\leq$, then the following notation and terminology applies:

- $a < b \iff a$ is less than $b \iff b > a \iff b$ is greater than a.
- $a \leq b \iff a$ is less than or equal to $b \iff b \geq a \iff b$ is greater than or equal to a.

We say A is **orderwise dense** if for each $a, b \in A$ with $a < b$, there exists another element $x \in A$ such that $a < x < b$.

Example 1.16. Let $A = \{a, b, c, d\}$ where a, b, c, d are distinct elements. Let P be the predicate

$$P(x) \equiv x = \langle a, b\rangle \vee x = \langle a, c\rangle \vee x = \langle a, d\rangle \vee x = \langle b, c\rangle \vee x = \langle b, d\rangle \vee x = \langle c, d\rangle.$$

By the existence of $A \times_K A$ and separation, we obtain the set $G_< = \{x \in A \times_K A \mid P(x)\}$, and we form $< = \langle G_<, \langle A, A\rangle\rangle$. Since $G_< \subseteq A \times_K A$, this is a relation. We may list out all the comparisons $<$ as follows:

$$\begin{array}{cccc}
a \not< a & a < b & a < c & a < d \\
b \not< a & b \not< b & b < c & b < d \\
c \not< a & c \not< b & c \not< c & c < d \\
d \not< a & d \not< b & d \not< c & d \not< d
\end{array}$$

As the reader may check directly, this relation was constructed specifically to satisfy the definition of a strict total order relation. Totally ordered sets are best visually represented by having their elements arranged in a line. For example, see the representation of A with $<$ below.

$$a \quad < \quad b \quad < \quad c \quad < \quad d$$

Figure 1.2: Visual representation of the total ordering relation on A.

Partial Order Relations

Here we introduce a generalization of total order relations by removing the property of (strict) comparability.

Definition. A **strict partial order relation** on A is a relation $<$ with the following properties for all $a, b, c \in A$:

(P1) Irreflexivity: $a \not< a$,

(P2) Asymmetry: if $a < b$ then $b \not< a$,

(P3) Transitivity: if $a < b$ and $b < c$, then $a < c$.

A **weak partial order relation** on A is a relation $\leq$ with the following properties for all $a, b, c \in A$:

(P1′) Reflexivity: $a \leq a$,

(P2′) Antisymmetry: if $a \leq b$ and $b \leq a$, then $a = b$,

(P3′) Transitivity: if $a \leq b$ and $b \leq c$, then $a \leq c$.

Given $<$ we may define a corresponding $\leq$ and vice-versa by declaring $\forall x \forall y\, (x \leq y \iff x < y \vee x = y)$ and $\forall x \forall y\, (x < y \iff x \leq y \wedge x \neq y)$.

If A has a strict or weak partial order relation, then we say A is a **partially ordered set** and introduce it as a pair $\langle A, < \rangle$. If A has a strict $<$ and a corresponding weak $\leq$, we write $b > a$ to denote $a < b$, and $b \geq a$ to denote $a \leq b$.

Example 1.17. Let $A = \{a, b, c, d, e, f\}$ where a, b, c, d, e, f are distinct elements. Let P be the predicate

$$P(x) \equiv x = \langle a, b \rangle \vee x = \langle a, c \rangle \vee x = \langle a, d \rangle \vee x = \langle a, e \rangle \vee x = \langle c, d \rangle \vee x = \langle c, e \rangle \vee x = \langle f, b \rangle.$$

By the existence of $A \times_K A$ and separation, we obtain the set $G_< = \{x \in A \times_K A \mid P(x)\}$, and we form $< = \langle G_<, \langle A, A \rangle \rangle$. Since $G_< \subseteq A \times_K A$, this is a relation. This gives us the following list:

$$a < b, \quad a < c, \quad a < d, \quad a < e, \quad c < d, \quad c < e, \quad f < b,$$

and $x \not< y$ for all cases not included above. As the reader may check directly, this relation was constructed specifically to satisfy the definition of a strict partial order relation.

Whereas total order relations are best represented by arranging elements in a line, partial order relations are best represented by arranging elements in a two-dimensional diagram called a **Hasse diagram**. In such a diagram, if x is below y and there is a path by line segments from x to y, then $x < y$. See the representation of $\langle A, < \rangle$ below.

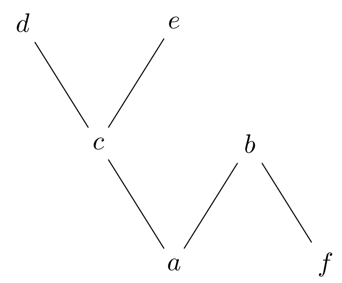

Figure 1.3: Hasse diagram of the partial ordering relation on A.

Below we present some terminology pertaining to partial orders and thus to total orders.

Definition. Let A be a partially ordered set with $<$ as the strict relation and $\leq$ as the corresponding weak relation, and let X be a subset of A. We define the notion of bounds.

- If an element $b \in A$ is such that $\forall x \in X\,(x \leq b)$, we say b is an **upper bound** of X. If an element $b \in A$ is such that $\forall x \in X\,(x < b)$, we say b is a **strict upper bound** of X. If an upper bound of X exists, we say X is **bounded above**.
- If an element $b \in A$ is such that $\forall x \in X\,(x \geq b)$, we say b is an **lower bound** of X. If an element $b \in A$ is such that $\forall x \in X\,(x > b)$, we say b is a **strict lower bound** of X. If an upper bound of X exists, we say X is **bounded below**.

We define the notions of maxima and minima.

- If $m \in X$ such that $\forall x \in X\,(x \leq m)$, then m is called the **maximum** or **greatest** element of X. If m exists, we write $m = \max X$.
- If $m \in X$ such that $\forall x \in X\,(m \leq x)$, then m is called the **minimum** or **least** element of X. If m exists, we write $m = \min X$.

We define notations of maximal and minimal elements.

- If $m \in X$ such that $\forall x \in X\,(m \leq x \implies x = m)$, then m is called a **maximal** element of X.
- If $m \in X$ such that $\forall x \in X\,(x \leq m \implies x = m)$, then m is called a **minimal** element of X.

In a totally ordered set, the greatest and maximal elements, if they exist, are the same thing. Likewise, in a totally ordered set, the least and minimal elements, if they exist, are the same thing. However, the concepts are not equivalent in general.

Example 1.18. As a simple example, consider $A = \{a, b, c\}$ with only the relationships $a < b$ and $a < c$. Here b and c are maximal, but neither b nor c is the greatest, because they are incomparable with one another. This gives us a concrete example where the concepts of maximal and greatest differ.

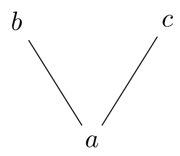

Figure 1.4: Hasse diagram of $\langle A, < \rangle$.

Proposition 1.19. *Let $\langle A, < \rangle$ be a partially ordered set. If $a, b \in A$ are the greatest elements of A, then $a = b$. If $c, d \in A$ are the least elements of A, then $c = d$.*

Proof. Suppose $a, b \in A$ are the greatest elements of A. By definition of greatest, $b \leq a$ and $a \leq b$, so $a = b$ by (P2′). Suppose $c, d \in A$ are the least elements of A. By definition of least, $c \leq d$ and $d \leq c$, so $c = d$ by (P2′). ∎

Let $\langle A, < \rangle$ be a partially ordered set. Proposition 1.19 shows that if the greatest element of A exists, it is unique. In contrast, maximal elements do not have to be unique, as shown in Example 1.18. Likewise, if the least element of A exists, it is unique, unlike minimal elements which do not have to be unique.

Equivalence Relations

Here we introduce another kind of relation.

Definition. An **equivalence relation** on A is a relation $\sim$ with the following properties for all $a, b, c \in A$:

(E1) Reflexivity: $a \sim a$,

(E2) Symmetry: if $a \sim b$ then $b \sim a$,

(E3) Transitivity: if $a \sim b$ and $b \sim c$, then $a \sim c$.

Example 1.20. Let $A = \{a, b, c, d, e\}$ where a, b, c, d, e are distinct elements. Define a relation $\sim$ on A by the following list:

$$\begin{array}{lllllllll} a \sim a, & a \sim b, & a \sim c, & b \sim a, & b \sim b, & b \sim c, & c \sim a, & c \sim b, & c \sim c, \\ d \sim d, & d \sim e, & e \sim d, & e \sim e, & & & & & \end{array}$$

and $x \not\sim y$ for all cases not included above. Justifying the existence of this relation follows the same process as in previous examples. We may verify this is an equivalence relation by checking all instances where $x \sim y$ and $x \not\sim y$ manually.

Example 1.21. Given any set A, use the existence of $A \times_K A$ and separation to obtain the set $G_E = \{\langle x, y\rangle \in A \times_K A \mid x = y\}$, and then form $E = \langle G_E, \langle A, A\rangle\rangle$. For all $x, y \in A$, we have $x \, E \, y$ if and only if $x = y$, so we have internalized the equality relation into a set-coded relation on A. It is easy to see E is an equivalence relation, and in this sense, equivalence relations are a generalization of the equality relation.

Sometimes we want to group various elements of a set A together. We do this as follows.

Definition. A **partition** of a set A is a collection $\mathcal{U}$ of subsets of A such that the following properties hold:

(a) No member of $\mathcal{U}$ is the empty set: $\forall x \in \mathcal{U}\,(x \neq \varnothing)$.
(b) The union of all members of $\mathcal{U}$ is A: $\bigcup \mathcal{U} = A$.
(c) The intersection of any two distinct members of $\mathcal{U}$ is the empty set: $\forall x, y \in \mathcal{U}\,(x \neq y \implies x \cap y = \varnothing)$.

Given an equivalence relation $\sim$ on set A, we can form a partition of A as follows. For each $a \in A$, use separation to obtain

$$\{a' \in A \mid a' \sim a\},$$

which we denote as $[a]_\sim$ or usually just $[a]$. Then use separation to obtain the set

$$\{x \in \mathcal{P}(A) \mid x = [a] \text{ for some } a \in A\},$$

which we will denote as $A/\sim$. We call $A/\sim$ the **quotient of** A **by** $\sim$ and we refer to elements of $A/\sim$ as **equivalence classes**. We leave it to the reader to show that for any $a, b \in A$ we have

$$a \sim b \iff [a] = [b], \tag{1.6}$$

and that $A/\sim$ is a partition of A. We may also go in the opposite direction. If $\mathcal{U}$ is a partition of A, we may define a relation $\sim$ on A by $a \sim b$ if and only if there exists a common class $C \in \mathcal{U}$ such that $a, b \in C$. We again leave it to the reader to show that the resulting relation is an equivalence relation.

Example 1.22. The quotient of $A = \{a, b, c, d, e\}$ by $\sim$ defined in Example 1.20 is

$$A/\sim = \{\{a, b, c\}, \{d, e\}\}.$$

This is a partition of A into subsets.

1.4. Set-Coded Functions

Basic Definitions

Based on the axioms telling us what is allowed and not allowed in set theory, we introduced various function-like operations like the power set operation $\mathcal{P}$, the union operation $\bigcup$, and the intersection operator $\bigcap$.

Just like with relations, we want to define functions only on certain sets, and we want functions to be treated as objects of set theory. For this reason, we will introduce new functions by encoding them as sets.

Definition. A **(set-coded) function** f from A to B is a relation $f = \langle G_f, \langle A, B\rangle\rangle$ in which the following two properties are satisfied:

(F1) Existence/Totality: For each $x \in A$, there exists $y \in B$ such that $\langle x, y\rangle \in G_f$.

(F2) Uniqueness: For all $x \in A$ and $y, y' \in B$, if $\langle x, y\rangle, \langle x, y'\rangle \in G_f$ then $y = y'$.

If f is a function, we define the following:

- the **domain** is $A = \mathrm{Dom}(f)$,
- the **codomain** is $B = \mathrm{Cod}(f)$,
- the **graph** is $G_f = \mathrm{Graph}(f)$.

We denote the fact that f is a function form A to B by writing $f : A \to B$.

Definition. A **partial function** f from A to B is a relation $f = \langle G_f, \langle A, B\rangle\rangle$ satisfying (F2) but not necessarily (F1). If f is a partial function, we define the following:

- the **ambient domain** is $A = \mathrm{AmbDom}(f)$,
- the **active domain** is $S = \mathrm{ActDom}(f)$ defined as the set of $x \in A$ such that $\langle x, y\rangle \in G_f$ for some $y \in B$,
- the **codomain** is $B = \mathrm{Cod}(f)$,
- the **graph** is $G_f = \mathrm{Graph}(f)$.

We denote the fact that f is a partial function from A to B by writing $f : A \rightharpoonup B$.

In our setup of terminology, a (set-coded) function is a partial function where the active domain is the ambient domain. We will oftentimes refer to a function as a **map** or **mapping**.

We think of a partial function as "sending" elements of the active domain to elements of the codomain. Given $f : A \rightharpoonup B$ with active domain S and $x \in S$, we want a way to automatically denote the unique $y \in B$ for which $\langle x, y\rangle \in G_f$. To this end, let D be the predicate

$$D(f, x) \equiv (f \text{ is a partial function}) \wedge x \in \mathrm{ActDom}(f),$$

and let P be the predicate

$$P(f,x,y) \equiv D(f,x) \wedge \langle x,y\rangle \in \mathrm{Graph}(f).$$

We leave it to the reader to verify

$$\forall f \forall x\,(D(f,x) \implies \exists! y\, P(f,x,y)).$$

Given this, we define a function symbol App with domain condition D such that $\mathrm{App}(f,x)$ is the unique y for which $P(f,x,y)$ holds, i.e. $\langle x,y\rangle \in \mathrm{Graph}(f)$. We abbreviate all well-formed terms of the form $\mathrm{App}(f,x)$ as simply $f(x)$. We may sometimes signify the fact that $y = f(x)$ by writing

$$x \mapsto y, \qquad f : x \mapsto y, \qquad x \overset{f}{\mapsto} y,$$

or some variation thereof.

Example 1.23. Let $A = \{a,b,c,d\}$ and $B = \{x,y,z\}$ where a,b,c,d are distinct elements of A and x,y,z are distinct elements of B. Let P be the predicate

$$P(t) \equiv t = \langle a,y\rangle \vee t = \langle b,x\rangle \vee t = \langle c,x\rangle \vee t = \langle d,x\rangle.$$

By the existence of $A \times_K B$ and separation, we obtain the set $G_f = \{t \in A \times_K B \mid P(t)\}$ and we form $f = \langle G_f, \langle A,B\rangle\rangle$.

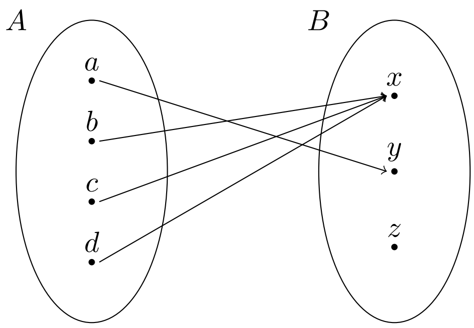

Figure 1.5: Visual representation of the function $f : A \to B$.

As one may check, properties (F1) and (F2) hold, so f is a function.

Example 1.24. Let $A = \{a,b,c,d\}$ and $B = \mathcal{P}(A)$ where a,b,c,d are distinct elements. Let P be the predicate

$$P(t) \equiv \exists x\,(x \in A \wedge t = \langle x, \{x\}\rangle).$$

By the existence of $A \times_K B$ and separation, we obtain the set $G_f = \{t \in A \times_K B \mid P(t)\}$ and we form $f = \langle G_f, \langle A,B\rangle\rangle$. As one may check, properties (F1) and (F2) hold, so f is a function.

Below we define some important sets that are associated with partial functions (which includes functions).

Definition. Let $f : A \rightharpoonup B$ be a partial function and let S be its active domain. We define the **image** of f to be

$$\operatorname{Im} f = \{y \in B \mid y = f(x) \text{ for some } x \in S\}.$$

If $E \subseteq A$, we define the **image of E under f** to be

$$f(E) = \{y \in B \mid y = f(x) \text{ for some } x \in E \cap S\}.$$

If $F \subseteq B$, we define the **inverse image of F under f** to be

$$f^{-1}(F) = \{x \in S \mid f(x) \in F\}.$$

Constructions justifying existence and uniqueness of these is done below.

Let D be the predicate

$$D(f, E) \equiv (f \text{ is a partial function}) \wedge E \subseteq \operatorname{AmbDom}(f),$$

and let P be the predicate

$$\begin{aligned} P(y, c) \equiv\ & (c \text{ is a Kuratowski ordered pair}) \\ & \wedge (\pi_L(c) \text{ is a partial function}) \\ & \wedge \pi_R(c) \subseteq \operatorname{AmbDom}(\pi_L(c)) \\ & \wedge \exists x\, (x \in \pi_R(c) \cap \operatorname{ActDom}(\pi_L(c)) \wedge y = \pi_L(c)(x)), \end{aligned}$$

which says c is a Kuratowski ordered pair consisting of a partial function f and a subset E of the ambient domain of f such that $y = f(x)$ for some $x \in E$. Then for all f, E such that $D(f, E)$ holds, by separation there exists a unique set of the form

$$\operatorname{Sep}_P(\operatorname{Cod}(f), \langle f, E\rangle),$$

where $\operatorname{Sep}_P(A, c)$ is the subset of A of elements y satisfying $P(y, c)$. We label this as $\operatorname{AppIm}(f, E)$, so AppIm is defined for all f, E for which $D(f, E)$ holds. Now $\operatorname{AppIm}(f, E)$ is precisely the image of E under f, and we abbreviate it as $f(E)$ as in the above definition. The image of f is now simply $\operatorname{Im} f = \operatorname{AppIm}(f, \operatorname{AmbDom}(f))$. The existence and uniqueness of inverse images can be shown similarly.

Example 1.25. Recall the function from Example 1.23 where $A = \{a, b, c, d\}$, $B = \{x, y, z\}$, and $f : A \to B$ is defined by $f(a) = y$, $f(b) = x$, $f(c) = x$, and $f(d) = x$. The codomain is $B = \{x, y, z\}$, but the image is $\operatorname{Im} f = \{x, y\}$.

Example 1.26. Recall the function from Example 1.24 where $A = \{a, b, c, d\}$, $B = \mathcal{P}(A)$, and $f : A \to B$ is defined by $f(x) = \{x\}$. The codomain is $\mathcal{P}(A)$, but the image is $\operatorname{Im} f = \{\{a\}, \{b\}, \{c\}, \{d\}\}$.

Given two sets, we may consider all possible functions from one to the other.

Definition. Given sets A and B, the set of all functions $f : A \to B$ from A to B is denoted by B^A or $\mathrm{Map}(A, B)$.

Proposition 1.27. *Given sets A and B, the set of all functions from A to B exists and is unique.*

Proof. First, we collect all graphs of functions from A to B into a set as follows. By taking the power set, we obtain $\mathcal{P}(A \times_K B)$. This is the collection of all sets $G \subseteq A \times_K B$. Let P be the predicate

$$\begin{aligned} P(G, c) \equiv (c \text{ is a Kuratowski ordered pair}) \wedge G \subseteq \pi_L(c) \times_K \pi_R(c) \\ \wedge (G \text{ satisfies (F1) and (F2) with respect to } \pi_L(c) \times_K \pi_R(c)). \end{aligned}$$

By separation, there exists a set

$$F = \{G \in \mathcal{P}(A \times_K B) \mid P(G, \langle A, B\rangle)\}.$$

This is the set of graphs of functions from A to B.

To proceed, we invoke the axiom schema of replacement (ZFC7) to attach the domain and codomain to the graphs. Let Q be the predicate

$$Q(G, f, c) \equiv f = \langle G, c\rangle.$$

Let us check that this is a functional predicate. Given G and c, we may form $f = \langle G, c\rangle$. If $Q(G, f, c)$ and $Q(G, \widetilde{f}, c)$, then $f, \widetilde{f}$ are both $\langle G, c\rangle$, so $f = \widetilde{f}$. Thus, Q is a functional predicate and by the axiom schema of replacement (ZFC7) with parameter $c = \langle A, B\rangle$, there exists a set $\widetilde{F} = \mathrm{Rep}_Q(F, \langle A, B\rangle)$ such that the following holds:

(i) If $f \in \widetilde{F}$, then there exists $G \in F$ such that $Q(G, f, \langle A, B\rangle)$, so $f = \langle G, \langle A, B\rangle\rangle$.
(ii) If $G \in F$, then there exists $f \in \widetilde{F}$ such that $Q(G, f, \langle A, B\rangle)$, so $f = \langle G, \langle A, B\rangle\rangle$.

So, given A, B, we took

$$\widetilde{F} = \mathrm{Rep}_Q(\mathrm{Sep}_P(\mathcal{P}(A \times_K B), \langle A, B\rangle), \langle A, B\rangle).$$

Next, we verify $\widetilde{F}$ is the set of all maps from A to B as intended. Suppose f is a function from A to B. By definition, $f = \langle G_f, \langle A, B\rangle\rangle$ such that $G_f \subseteq A \times_K B$ satisfies (F1) and (F2). We find $G_f \in F$, so by property (ii), there exists some $\widetilde{f} \in \widetilde{F}$ such that

$$\widetilde{f} = \langle G_f, \langle A, B\rangle\rangle.$$

The right-hand side is precisely f, so $f \in \widetilde{F}$. Conversely, suppose $f \in \widetilde{F}$. By property (i), there exists some $G \in F$ such that $f = \langle G, \langle A, B\rangle\rangle$. Given that $G \in F$, G satisfies (F1) and (F2), so f is a function.

Thus, f is a function from A to B if and only if $f \in \widetilde{F}$. The uniqueness of the set of all functions from A to B follows straightforwardly by extensionality. ■

Compositions and Changes of Domain and Codomain

We will be using the following useful lemma.

Lemma 1.28. *Suppose $f, g : A \to B$ are functions with the same domain A and codomain B. If $f(x) = g(x)$ for all $x \in A$, then $f = g$.*

Proof. Write $f = \langle G_f, \langle A, B\rangle\rangle$ and $g = \langle G_g, \langle A, B\rangle\rangle$. We aim to show the graphs G_f, G_g are equal. Given $t \in G_f$, we know $t = \langle x, f(x)\rangle$ for some $x \in A$. Since $f(x) = g(x)$ by hypothesis, substitution implies $t = \langle x, g(x)\rangle$, and $\langle x, g(x)\rangle \in G_g$. Thus, by substitution, $t \in G_g$. Conversely, if $t \in G_g$, analogous reasoning shows $t \in G_f$. By extensionality, $G_f = G_g$ and so $f = g$. ∎

We introduce composition of functions.

Definition (Composition)**.** Let $f : A \to B$ and $g : B \to C$ be functions. We define the **composition** of these two functions to be the function $g \circ f : A \to C$ such that $g \circ f(a) = g(f(a))$ for all $a \in A$. We verify $g \circ f$ exists and is unique below.

Proposition 1.29 (Existence and uniqueness of compositions)**.** *If $f : A \to B$ and $g : B \to C$ are functions, then the composition $g \circ f$ exists and is unique.*

Proof. Write $f = \langle G_f, \langle A, B\rangle\rangle$ and $g = \langle G_g, \langle B, C\rangle\rangle$. By the existence of $A \times_K C$ and by separation, there exists the set

$$G = \{t \in A \times_K C \mid t = \langle x, g(f(x))\rangle \text{ for some } x \in A\}.$$

More formally, this is

$$G = \{t \in A \times_K C \mid P(t, \langle f, g\rangle)\}$$

where

$$\begin{aligned} P(t, c) \equiv\ & (c \text{ is a Kuratowski ordered pair}) \wedge (\pi_L(c), \pi_R(c) \text{ are functions}) \\ & \wedge \operatorname{Cod}(\pi_L(c)) = \operatorname{Dom}(\pi_R(c)) \\ & \wedge \exists x\, (x \in \operatorname{Dom}(\pi_L(c)) \wedge t = \langle x, \operatorname{App}(\pi_R(c), \operatorname{App}(\pi_L(c), x))\rangle). \end{aligned}$$

Now define $h = \langle G, \langle A, C\rangle\rangle$, which exists. Since $G \subseteq A \times_K C$, it remains to verify (F1) and (F2).

For (F1), for each $x \in A$, we have $\langle x, g(f(x))\rangle \in A \times_K C$ and $P(\langle x, g(f(x))\rangle, \langle f, g\rangle)$, so

$$\langle x, g(f(x))\rangle \in G,$$

so (F1) holds. For (F2), suppose $x \in A$ and $z, z' \in C$ such that

$$\langle x, z\rangle, \langle x, z'\rangle \in G.$$

By definition of G, there exist $u, u' \in A$ such that

$$\langle x, z\rangle = \langle u, g(f(u))\rangle \quad \text{and} \quad \langle x, z'\rangle = \langle u', g(f(u'))\rangle.$$

By (1.4), $x = u$, $z = g(f(u))$, $x = u'$, and $z' = g(f(u'))$. As a result we have $u = u'$, so by substitution, $g(f(u)) = g(f(u'))$, and so $z = z'$ and (F2) holds.

This shows $h : A \to C$ is a function such that $h(x) = g(f(x))$ for all $x \in A$ by construction. Uniqueness follows from Lemma 1.28. ■

By existence and uniqueness of $g \circ f$ given composable functions f, g, we may create a composition formation operator as follows. Take the domain condition D and predicate P so that

$$D(f, g) \equiv (f, g \text{ are functions}) \land \mathrm{Cod}(f) = \mathrm{Dom}(g)$$

and

$$\begin{aligned} P(f, g, h) \equiv D(f, g) \land (h \text{ is a function}) \land \mathrm{Dom}(h) = \mathrm{Dom}(f) \\ \land \mathrm{Cod}(h) = \mathrm{Cod}(g) \land \forall x \in \mathrm{Dom}(f)\,(h(x) = g(f(x))). \end{aligned}$$

Given

$$\forall f \forall g\,(D(f, g) \implies \exists! h\, P(f, g, h)),$$

we obtain the operation Comp with domain condition D such that $h = \mathrm{Comp}(f, g)$ is the composition. We abbreviate $\mathrm{Comp}(f, g)$ as $g \circ f$.

We introduce changes of domains and codomains of functions.

Definition (Change of domain and codomain)**.** Let $f : A \to B$, $X \subseteq A$, and $Y \supseteq f(X)$. We define $f|_X^Y$ to be the function from X to Y such that $f|_X^Y(x) = f(x)$ for all $x \in X$. Notation for special cases:

- If $Y = B$ (the codomain isn't changed), we write $f|_X^B = f|_X$.
- If $X = A$ (the domain isn't changed), we write $f|_A^Y = f|^Y$.

We verify $f|_X^Y$ exists and is unique below.

Proposition 1.30 (Existence and uniqueness of change of domain and codomain)**.** *If $f : A \to B$ is a function, $X \subseteq A$, and $Y \supseteq f(X)$, then the function with a change of domain and codomain $f|_X^Y$ exists and is unique.*

Proof. Given $f = \langle G_f, \langle A, B\rangle\rangle$, $X \subseteq A$, and $Y \supseteq f(X)$, use separation to obtain the subset

$$G = \{t \in G_f \mid t \in X \times_K Y\}.$$

More formally, this is

$$G = \{t \in G_f \mid P(t, \langle X, Y\rangle)\}$$

where

$$P(t, c) \equiv (c \text{ is a Kuratowski ordered pair}) \land t \in \pi_L(c) \times_K \pi_R(c).$$

Now define $h = \langle G, \langle X, Y\rangle\rangle$, which exists. If $t \in G$, then $P(t, \langle X, Y\rangle)$, so $t \in X \times_K Y$. Hence $G \subseteq X \times_K Y$, and it remains to verify (F1) and (F2).

For (F1), for each $x \in X$, we have $\langle x, f(x)\rangle \in G_f$ and $f(x) \in f(X) \subseteq Y$, and so $P(\langle x, f(x)\rangle, \langle X, Y\rangle)$. Thus,

$$\langle x, f(x)\rangle \in G,$$

so (F1) holds. For (F2), suppose $x \in X$ and $y, y' \in Y$ such that

$$\langle x, y\rangle, \langle x, y'\rangle \in G.$$

Since $G \subseteq G_f$ and G_f has property (F2), it follows $y = y'$, and so (F2) holds.

This shows $h : X \to Y$ is a function such that $h(x) = f(x)$ for all $x \in X$ by construction. Uniqueness follows from Lemma 1.28. ■

By existence and uniqueness of $f|_X^Y$ given $f : A \to B$, $X \subseteq A$, and $Y \supseteq f(X)$, we may create a domain/codomain change operator as follows. Take the domain condition D and predicate P so that

$$\begin{aligned} D(f,p) \equiv{} & (f \text{ is a function}) \wedge (p \text{ is a Kuratowski ordered pair}) \\ & \wedge \pi_L(p) \subseteq \mathrm{Dom}(f) \wedge \pi_R(p) \supseteq f(\pi_L(p)) \end{aligned}$$

and

$$\begin{aligned} P(f,p,h) \equiv{} & D(f,p) \wedge (h \text{ is a function}) \wedge \mathrm{Dom}(h) = \pi_L(p) \\ & \wedge \mathrm{Cod}(h) = \pi_R(p) \wedge \forall x \in \pi_L(p)\, (f(x) = h(x)). \end{aligned}$$

Given

$$\forall f \forall p\, (D(f,p) \implies \exists! h\, P(f,p,h)),$$

we obtain the operation Change with domain condition D such that $h = \mathrm{Change}(f,p)$ is the function obtained from f by restricting the domain to $X = \pi_L(p)$ and changing the codomain to $Y = \pi_R(p)$. We abbreviate $\mathrm{Change}(f, \langle X, Y\rangle)$ as $f|_X^Y$.

Identity Maps and Inverses

Now we present some ubiquitous types of functions.

Definition (The identity map)**.** Given a set A, we define the **identity map** of A to be the function denoted $\mathrm{id}_A : A \to A$ that outputs exactly the same element that is given as the input: $\mathrm{id}_A(x) = x$ for all $x \in A$.

We leave it to the reader to construct the identity map and prove uniqueness in the ending problem set.

Definition. Let $f : A \to B$ and $g : B \to A$ be maps.

- g is said to be a **left-inverse** of f if $g \circ f = \mathrm{id}_A$.
- g is said to be a **right-inverse** of f if $f \circ g = \mathrm{id}_B$.

- g is said to be an **inverse** of f if it is both a left-inverse and right-inverse. We show below that if it exists, then it is unique. We denote it by f^{-1}, which can be done due to uniqueness.

If an inverse of f exists, f is called **invertible**.

Proposition 1.31. *If $f : A \to B$ is an invertible function, then its left-inverse and right-inverse are unique and the left-inverse and right-inverse are equal to each other.*

Proof. Let $f : A \to B$ be an invertible function and let $g : B \to A$ be its inverse. Suppose $g_L : B \to A$ is a left-inverse of f. For any $y \in B$, we have $f(g(y)) = y$ because g is an inverse of f, so $x = g(y)$ is the element such that $f(x) = y$. Then because g, g_L are left-inverses of f,

$$g_L(f(x)) = x = g(f(x)), \quad \text{so} \quad g_L(y) = g(y).$$

By Lemma 1.28, $g_L = g$, which means all left-inverses of f are g.

Suppose $g_R : B \to A$ is a right-inverse of f. For any $y \in B$, we have $f(g(y)) = y = f(g_R(y))$ because g, g_R are right-inverses of f, so

$$g(f(g(y))) = g(f(g_R(y))), \quad \text{so} \quad g(y) = g_R(y),$$

because g is an inverse of f. By Lemma 1.28, $g_R = g$, which means all right-inverses of f are g.

Lastly, for any left-inverse g_L and right-inverse g_R of f, both are equal to g, so $g_L = g_R$. ■

Example 1.32 (Left-inverse)**.** Let $A = \{a, b, c\}$ and $B = \{x, y, z, w\}$ where a, b, c, x, y, z, w are distinct elements. Define $f : A \to B$ by $f(a) = x$, $f(b) = y$, and $f(c) = z$. Define $g : B \to A$ by $g(x) = a$, $g(y) = b$, $g(z) = c$, and $g(w) = c$. Then

$$g(f(a)) = a, \quad g(f(b)) = b, \quad \text{and} \quad g(f(c)) = c,$$

so $g \circ f = \mathrm{id}_A$ and g is a left-inverse of f. However, $f(g(w)) = z \neq w$, so g is not a right-inverse of f.

Example 1.33 (Right-inverse)**.** Let $A = \{a, b, c, d\}$ and $B = \{x, y, z\}$ where a, b, c, d, x, y, z are distinct elements. Define $f : A \to B$ by $f(a) = x$, $f(b) = y$, $f(c) = z$, and $f(d) = z$. Define $g : B \to A$ by $g(x) = a$, $g(y) = b$, and $g(z) = c$. Then

$$f(g(x)) = x, \quad f(g(y)) = y, \quad \text{and} \quad f(g(z)) = z,$$

so $f \circ g = \mathrm{id}_B$ and g is a right-inverse of f. However, $g(f(d)) = c \neq d$, so g is not a left-inverse of f.

Example 1.34 (Inverse)**.** Let $A = \{a, b, c\}$ and $B = \{x, y, z\}$ where a, b, c, x, y, z are distinct elements. Define $f : A \to B$ by $f(a) = y$, $f(b) = z$, and $f(c) = x$. Define $g : B \to A$ by $g(x) = c$, $g(y) = a$, and $g(z) = b$. Then

$$\begin{aligned} &g(f(a)) = a, \qquad &&g(f(b)) = b, \qquad &&g(f(c)) = c, \\ &f(g(x)) = x, &&f(g(y)) = y, &&f(g(z)) = z, \end{aligned}$$

so $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$. Thus, g is an inverse of f.

Injective, Surjective, and Bijective Maps

Definition. Let $f : A \to B$ be a function.

- We say f is **injective** if for all $x, y \in A$ we have $f(x) = f(y) \implies x = y$.
- We say f is **surjective** if for all $y \in B$ there exists $x \in A$ such that $f(x) = y$.
- We say f is **bijective** if it is both injective and surjective.

An equivalent characterization of injective functions is that $f : A \to B$ is injective if and only if for all $x, y \in A$ such that $x \neq y$ we have $f(x) \neq f(y)$. That is, injective functions send distinct elements to distinct elements. An equivalent characterization of surjective functions is that $f : A \to B$ is surjective if and only if $\operatorname{Im} f = B$. That is, surjective functions are precisely the ones in which their image matches their codomain. We provide some illustrations of some examples below.

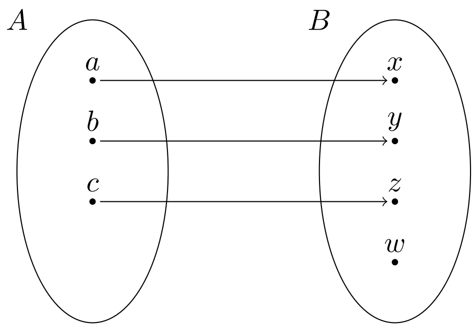

Figure 1.6: An injective but not surjective function.

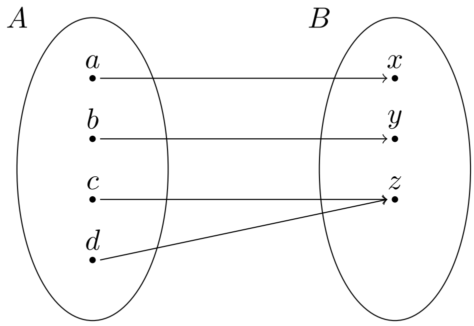

Figure 1.7: A surjective but not injective function.

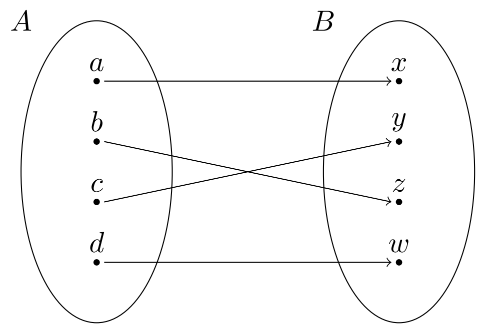

Figure 1.8: A bijective function.

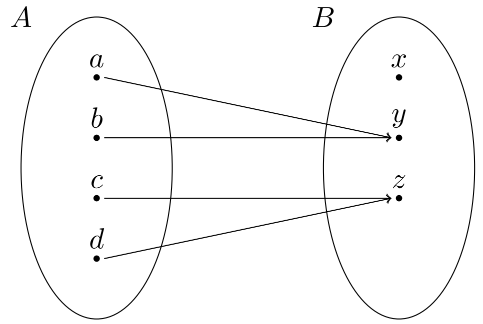

Figure 1.9: A function that is neither injective nor surjective.

Functions and Quotients

Let us revisit the notion of equivalence classes and quotient sets. Given a set A and an equivalence relation $\sim$ on A, the quotient set $A/\sim$ is defined as

$$A/\sim = \{x \in \mathcal{P}(A) \,|\, x = [a] \text{ for some } a \in A\} \quad \text{where} \quad [a] = \{a' \in A \,|\, a' \sim a\}$$

for each $a \in A$. We often want to define functions on quotient sets, which would be of the form $f : A/\sim \to B$. However, we have to be careful: If we define $f([a])$ in terms of a representative a of the equivalence class $[a]$, we must ensure that the resulting value is independent of the choice of representative. If $a \sim a'$, then the rule used to define $f([a])$ must assign the same value whether we started with representative a or representative a'. If this does not hold, then in informal treatments we say the function is **not well-defined**. Formally speaking, this actually means the function simply does not exist.

Example 1.35 (f is not well-defined)**.** Let $A = \{a, b, c, d, e, i\}$ and $B = \{\text{Consonant}, \text{Vowel}\}$ where all elements listed are distinct from one another. Let $P(t) \equiv t = a \vee t = e \vee t = i$. Define a relation $\sim$ on A by

$$a \sim a, \qquad b \sim b, \qquad c \sim c, \qquad d \sim d, \qquad e \sim e, \qquad i \sim i,$$

$$a \sim b, \qquad b \sim a, \qquad c \sim d, \qquad d \sim c, \qquad e \sim i, \qquad i \sim e.$$

We leave it to the reader to check this is an equivalence relation on A. The resulting quotient set is

$$A/\sim\, = \{\{a, b\}, \{c, d\}, \{e, i\}\}.$$

Define $f : A/\sim\, \to B$ as follows:

$$f([x]) = \begin{cases} \text{Vowel} & \text{if } P(x), \\ \text{Consonant} & \text{otherwise.} \end{cases}$$

This function is not well-defined! The problem is that because $[a] = [b]$, it must be the case that $f([a]) = f([b])$, but if we follow our naive rule then $f([a]) =$ Vowel and $f([b]) =$ Consonant.

On one hand, $f([a]) = f([b])$ but we declared at the start that Consonant and Vowel are distinct elements: Consonant $\neq$ Vowel. The mere introduction of this function drives us into a contradiction, so it must not exist. This is where it is necessary to fall back to set-theoretic constructions as in Example 1.23 and Example 1.24.

Below, we provide the appropriate definitions and theorems needed to describe a fool-proof recipe for defining functions on quotient sets.

Definition. If $\sim$ is an equivalence relation on a set A, the map $q : A \to A/\sim$ defined by $q(a) = [a]$ is called the **quotient map**.

Here is our construction for the above definition. Given any relation $R = \langle G_R, \langle A, B\rangle\rangle$, we take $\mathrm{Dom}(R) = A$ and $\mathrm{Cod}(R) = B$. Take $A \times_K A/\sim$ and apply separation to get

$$G_q = \{t \in A \times_K A/\sim\, \mid P(t, \sim)\}$$

where

$$P(t, \sim) \equiv (\sim \text{ is an equivalence relation}) \wedge \exists a\, (a \in \mathrm{Dom}(\sim) \wedge t = \langle a, [a]_\sim\rangle).$$

Here $G_q \subseteq A \times_K A/\sim$ and the reader may verify that it satisfies (F1) and (F2). By forming $q = \langle G_q, \langle A, A/\sim\rangle\rangle$, we verify that q indeed exists as a function.

Lemma 1.36. *If $\sim$ is an equivalence relation on a set A, then the quotient map $q : A \to A/\sim$ is surjective.*

Proof. If $x \in A/\sim$ then $x = [a]$ for some $a \in A$, so $q(a) = x$. ■

Now we describe the main strategy for defining functions on quotient sets. Given a set A with an equivalence relation $\sim$, instead of trying to define a map $g : A/\sim\, \to B$ directly on domain $A/\sim$ (which may result in an inconsistent definition like in Example 1.35 if we are not careful), it makes sense to define a map $f : A \to B$ on domain A, and if this map f meets certain conditions, it passes to a corresponding map $g : A/\sim\, \to B$ guaranteed to be well-defined. This is justified and made precise by the theorem below.

Theorem 1.37 (Quotient factorization theorem for sets)**.** *Suppose $\sim$ is an equivalence relation on a set A and $f : A \to B$ is a function such that $a \sim a'$ implies $f(a) = f(a')$ for all $a, a' \in A$. Then there exists a unique function $g : A/\sim \to B$ such that*

$$f = g \circ q \tag{1.7}$$

where q is the quotient map with respect to $\sim$.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "q"'] & B \\ A/\sim \arrow[ur, dashed, "g"'] & \end{tikzcd}$$

Figure 1.10: A commutative diagram illustrating the quotient factorization theorem for sets. A **commutative diagram** is a diagram of objects and functions in which any two paths with the same start and end points yield the same output for each input.

Proof. EXISTENCE. Let $Q = A/\sim$, let $q : A \to Q$ be the quotient map, and form the product $Q \times_K B$. By separation, take the set

$$G_g = \{t \in Q \times_K B \mid t = \langle [a], f(a) \rangle \text{ for some } a \in A\}.$$

Formally, this is

$$G_g = \{t \in Q \times_K B \mid P(t, \langle f, \sim \rangle)\}$$

where

$$\begin{aligned} P(t, c) \equiv {} & (c \text{ is a Kuratowski ordered pair}) \wedge (\pi_L(c) \text{ is a function}) \\ & \wedge (\pi_R(c) \text{ is an equivalence relation}) \\ & \wedge \exists a \, (a \in \mathrm{Dom}(\pi_L(c)) \wedge t = \langle [a]_{\pi_R(c)}, \pi_L(c)(a) \rangle). \end{aligned}$$

Now form $g = \langle G_g, \langle Q, B \rangle \rangle$, which exists. If $t \in G_g$, then $t = \langle [a], f(a) \rangle$ for some $a \in A$, so $t \in Q \times_K B$. Thus, $G_g \subseteq Q \times_K B$, and it remains to show it satisfies (F1) and (F2).

For (F1), let $x \in Q$. By definition of quotients, there exists $a \in A$ such that $x = [a]$. Given that $f(a) \in B$,

$$\langle x, f(a) \rangle = \langle [a], f(a) \rangle \in Q \times_K B,$$

and also $P(\langle x, f(a) \rangle, \langle f, \sim \rangle)$ holds, so $\langle x, f(a) \rangle \in G_g$. Thus, (F1) holds.

For (F2), suppose $x \in Q$ and $y, y' \in B$ such that

$$\langle x, y \rangle, \langle x, y' \rangle \in G_g.$$

By definition of G_g, there exist $a, a' \in A$ such that

$$\langle x, y\rangle = \langle [a], f(a)\rangle \quad \text{and} \quad \langle x, y'\rangle = \langle [a'], f(a')\rangle.$$

By (1.4),

$$x = [a], \quad y = f(a), \quad x = [a'], \quad \text{and} \quad y' = f(a').$$

So $[a] = [a']$, which implies $a \sim a'$ by (1.6). By hypothesis of f, it follows $f(a) = f(a')$ and so $y = y'$. Thus, (F2) holds. This proves $g : Q \to B$ exists and is a function.

By construction, $g \circ q(a) = g(q(a)) = g([a]) = f(a)$ for all $a \in A$, so by Lemma 1.28, (1.7) holds.

Uniqueness. Suppose $g, g' : Q \to B$ are two functions such that $f = g \circ q$ and $f = g' \circ q$. By hypothesis, g and g' have the same domain Q and codomain B. Given $x \in Q$, by construction of Q we know $x = [a]$ for some $a \in A$. Then

$$g(x) = g([a]) = g(q(a)) = f(a) = g'(q(a)) = g'([a]) = g'(x).$$

This applies to all $x \in Q$, so by Lemma 1.28, it follows $g = g'$. ■

Example 1.38. Let $A = \{a, b, c, \delta, \varepsilon\}$ and $B = \{\text{Latin}, \text{Greek}\}$ where all elements listed are distinct. Let $P(t) \equiv t = a \vee t = b \vee t = c$. Define a relation $\sim$ on A by

$$\begin{aligned} &a \sim a, \quad b \sim b, \quad c \sim c, \quad a \sim b, \quad b \sim a, \quad a \sim c, \quad c \sim a, \quad b \sim c, \quad c \sim b, \\ &\delta \sim \delta, \quad \varepsilon \sim \varepsilon, \quad \delta \sim \varepsilon, \quad \varepsilon \sim \delta. \end{aligned}$$

We leave it to the reader to check this is an equivalence relation on A. The resulting quotient is

$$A/\sim\, = \{\{a, b, c\}, \{\varepsilon, \delta\}\}.$$

Instead of defining a function from $A/\sim$ to B directly like we tried in Example 1.35, we follow the more principled approach by defining $f : A \to B$ by

$$f(x) = \begin{cases} \text{Latin} & \text{if } P(x), \\ \text{Greek} & \text{otherwise.} \end{cases}$$

Let us construct this function to ensure it exists. Apply separation to $A \times_K B$ to obtain

$$G_f = \{t \in A \times_K B \mid t = \langle x, \text{Latin}\rangle \text{ if } P(x) \text{ and } t = \langle x, \text{Greek}\rangle \text{ otherwise}\}.$$

More formally, this is $G_f = \{t \in A \times_K B \mid Q(t)\}$ where

$$Q(t) \equiv \exists x\, ((P(x) \implies t = \langle x, \text{Latin}\rangle) \wedge (\neg P(x) \implies t = \langle x, \text{Greek}\rangle)).$$

We form $f = \langle G_f, \langle A, B\rangle\rangle$ and leave the reader to verify that $G_f \subseteq A \times_K B$, (F1), and (F2) hold, proving f exists and is a function. Now because

$$f(a) = f(b) = f(c) = \text{Latin} \quad \text{and} \quad f(\delta) = f(\varepsilon) = \text{Greek},$$

the relationship $x \sim x'$ implies $f(x) = f(x')$ for all $x, x' \in A$, and so Theorem 1.37 is applicable. Thus, there exists a unique function $g : A/\sim \to B$ such that $f = g \circ q$ where q is the quotient map. In this case,

$$g(\{a, b, c\}) = \text{Latin} \quad \text{and} \quad g(\{\delta, \varepsilon\}) = \text{Greek}.$$

In more loose terms, we may say $g([x]) = \text{Latin}$ if $P(x)$ and $g([x]) = \text{Greek}$ if $\neg P(x)$ without running into trouble, because Theorem 1.37 guarantees this leaves g as a well-defined map.

1.5. Choice Principles

The Axiom of Choice

We introduce the following axiom.

(ZFC9) Axiom of Choice. Given a set A consisting of nonempty sets, there exists a function

$$f : A \to \bigcup A$$

such that $f(a) \in a$ for all $a \in A$.

Any function matching the description of f in the axiom of choice is called a **choice function**. The meaning of such a function is that for each $a \in A$, the function picks out some element of a. A very important fact about the axiom of choice is that it is making an assertion about arbitrary collections of nonempty sets. Let us go over an application of the axiom of choice.

Proposition 1.39. *Let $f : A \to B$ be a function whose domain A is nonempty. Then the following statements are true:*

(a) *f is injective if and only if it has a left-inverse. Every left-inverse, if any exists, is surjective.*
(b) *f is surjective if and only if it has a right-inverse. Every right-inverse, if any exists, is injective.*
(c) *f is bijective if and only if it has an inverse. The inverse, if it exists, is bijective.*

Remark 1.40. Part (b) of the proposition specifically requires the axiom of choice. In the ending problem set, we claim that (b) is actually equivalent to the axiom of choice. Recall from Proposition 1.31 that the inverse of f, if it exists, is unique.

Proof. (a). Suppose f is injective. Since A is nonempty, we may take an arbitrary element $a_0 \in A$. We define $g : B \to A$ by

$$g(y) = \begin{cases} x & \text{if } y \in \operatorname{Im} f, \text{ where } x \text{ is the unique element with } f(x) = y, \\ a_0 & \text{if } y \notin \operatorname{Im} f. \end{cases}$$

The element x in the first case is unique precisely because f is injective. To justify existence, apply separation to $B \times_K A$ to obtain

$$G_g = \{o \in B \times_K A \,|\, P(o, \langle f, a_0 \rangle)\}$$

where

$$\begin{aligned} P(o,c) \equiv{}& (o, c \text{ are Kuratowski ordered pair}) \wedge (\pi_L(c) \text{ is a function}) \\ & \wedge (\underbrace{\langle \pi_R(o), \pi_L(o)\rangle \in \mathrm{Graph}(\pi_L(c))}_{\text{first disjunct}} \vee \underbrace{(\pi_L(o) \notin \mathrm{Im}(\pi_L(c)) \wedge \pi_R(o) = \pi_R(c))}_{\text{second disjunct}}). \end{aligned}$$

Reading $o = \langle y, x\rangle$, the predicate says either $f(x) = y$, or $y \notin \operatorname{Im} f$ and $x = a_0$. Take $g = \langle G_g, \langle B, A\rangle\rangle$, which exists. Since $G_g \subseteq B \times_K A$, it remains to verify (F1) and (F2) from the definition of functions.

For (F1), let $y \in B$. If $y \in \operatorname{Im} f$, there exists $x \in A$ with $f(x) = y$, so $\langle x, y\rangle \in \mathrm{Graph}(f)$, and so $\langle y, x\rangle$ satisfies the first disjunct and $\langle y, x\rangle \in G_g$. Else if $y \notin \operatorname{Im} f$, then $\langle y, a_0\rangle$ satisfies the second disjunct and $\langle y, a_0\rangle \in G_g$. In both cases, there exists $x \in A$ such that $\langle y, x\rangle \in G_g$.

For (F2), suppose $\langle y, x\rangle, \langle y, x'\rangle \in G_g$. If $y \in \operatorname{Im} f$, the second disjunct fails for both pairs, so $\langle x, y\rangle, \langle x', y\rangle \in \mathrm{Graph}(f)$, i.e. $f(x) = y = f(x')$, and injectivity of f gives $x = x'$. Else if $y \notin \operatorname{Im} f$, the first disjunct fails for both pairs, so $x = a_0 = x'$. In both cases, $x = x'$.

Thus, $g : B \to A$ is a function with $g(y) = x$ whenever $f(x) = y$. Given any $x \in A$, take $y = f(x)$, so then $y \in \operatorname{Im} f$ and the unique element of A mapping to y under f is x, so

$$g \circ f(x) = g(f(x)) = g(y) = x.$$

Hence $g \circ f = \mathrm{id}_A$ by Lemma 1.28, and g is a left-inverse of f.

Conversely, suppose f has a left-inverse $g : B \to A$. Then if $x, x' \in A$ such that $f(x) = f(x')$, we find that by applying g to both sides, we get $g(f(x)) = g(f(x'))$, so

$$x = g \circ f(x) = g(f(x)) = g(f(x')) = g \circ f(x') = x',$$

and so f is injective.

Lastly, suppose f is injective and $g : B \to A$ is any left-inverse. For any $x \in A$, we have $g(y) = x$ for $y = f(x)$, so g is surjective.

(b). Suppose f is surjective. Let $i : B \to \mathcal{P}(A)$ by $i(y) = f^{-1}\{y\}$ and $C = \operatorname{Im} i$. To justify existence, apply separation to $B \times_K \mathcal{P}(A)$ to obtain

$$G_i = \{o \in B \times_K \mathcal{P}(A) \,|\, Q(o, f)\}$$

where

$$\begin{aligned} Q(o,c) \equiv{}& (o \text{ is a Kuratowski ordered pair}) \wedge (c \text{ is a function}) \\ & \wedge (\pi_L(o) \in \mathrm{Cod}(c)) \wedge (\pi_R(o) = c^{-1}\{\pi_L(o)\}). \end{aligned}$$

Given $G_i \subseteq B \times_K \mathcal{P}(A)$, we verify (F1) and (F2) from the definition of functions. Here (F1) holds because for all $y \in B$ we have $\langle y, f^{-1}\{y\}\rangle \in G_i$, and (F2) holds because if $\langle y, X\rangle, \langle y, X'\rangle \in G_i$ then $X = f^{-1}\{y\} = X'$.

By surjectivity of f, every inverse image of the form $f^{-1}\{y\}$ for $y \in B$ is nonempty, so every member of C is nonempty. By the axiom of choice (ZFC9), there exists a function $j : C \to \bigcup C$ such that $j(c) \in c$ for all $c \in C$. Note that since $C \subseteq \mathcal{P}(A)$, we have $\bigcup C \subseteq A$. Then we may form

$$g = j|^A \circ i|^C : B \to A.$$

For any $y \in B$, we have

$$f \circ g(y) = f(g(y)) = f(j(f^{-1}\{y\})) = y$$

where the last equality follows from $j(f^{-1}\{y\}) \in f^{-1}\{y\}$ and $\forall x\,(x \in f^{-1}\{y\} \implies f(x) = y)$. Hence $f \circ g = \mathrm{id}_B$ by Lemma 1.28, and g is a right-inverse of f.

Conversely, suppose f has a right-inverse $g : B \to A$. Then for any $y \in B$, we have $f(x) = y$ for $x = g(y)$, so f is surjective.

Lastly, suppose f is surjective and $g : B \to A$ is any right-inverse. If $y, y' \in B$ such that $g(y) = g(y')$, then $y = f(g(y)) = f(g(y')) = y'$, so g is injective.

(c). Suppose f is bijective, meaning it is both injective and surjective. By (a), f has a left-inverse $g_L : B \to A$, and by (b), f has a right-inverse $g_R : B \to A$. We claim $g_L = g_R$. For any $y \in B$, applying the left-inverse identity $g_L \circ f = \mathrm{id}_A$ to $g_R(y)$ gives

$$g_L(f(g_R(y))) = g_R(y).$$

Since $f \circ g_R = \mathrm{id}_B$, we have $f(g_R(y)) = y$, so the left-hand side above is $g_L(y)$. Thus, $g_L(y) = g_R(y)$, and since this holds for all $y \in B$, Lemma 1.28 gives $g_L = g_R$. Setting $g = g_L = g_R$, we find $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$, so g is an inverse of f.

Conversely, suppose f has an inverse g. Then g is both a left-inverse and a right-inverse of f, so by (a) and (b), f is injective and surjective. Thus, f is bijective.

Lastly, if g is an inverse of f, then $f \circ g = \mathrm{id}_B$ and $g \circ f = \mathrm{id}_A$ show f is an inverse of g, so g is invertible. Applying the forward direction just proved to g in place of f, we conclude g is bijective. ■

Zorn's Lemma

Let us provide an important definition about partial orders.

Definition. Let $\langle P, <\rangle$ be a partially ordered set. Suppose C is a subset of P and $<'$ is the restriction of $<$ to C. If $\langle C, <'\rangle$ is a totally ordered set, we call it a **chain** of $\langle P, <\rangle$. For brevity, we usually refer to $\langle P, <\rangle$ and $\langle C, <'\rangle$ as just P and C.

A famous consequence of the axiom of choice involving partial orders is Zorn's lemma. Zorn's lemma is in fact equivalent to the axiom of choice (given all the other axioms), but for the purposes of our text we are concerned with only one implication direction.

Theorem 1.41 (Zorn's lemma)**.** *Let $\langle P, < \rangle$ be a nonempty partially ordered set. If every chain C of P is bounded above, then there exists a maximal element of P.*

Proof. Let $\langle P, < \rangle$ be a nonempty partially ordered set and suppose the following:

(i) Every chain of P is bounded above: If $C \subseteq P$ is a chain, there exists $y \in P$ such that $x \leq y$ for all $x \in C$.
(ii) There is no maximal element of P: If $m \in P$, then there exists $x \in P$ such that $m < x$.

We will show that these two facts lead to a contradiction, so both cannot be true at the same time. Thus, if a nonempty partially ordered set satisfies (i), then the set must have a maximal element, which means Zorn's lemma is true. Let $\mathcal{C} \subseteq \mathcal{C}(P)$ be the collection of all chains of P. For any chain C of P, define

$$C_x = \{y \in C \mid y < x\}.$$

Suppose $C \subseteq P$ is a chain. By fact (i), there exists $m \in P$ such that $x \leq m$ for all $x \in C$. By fact (ii), there exists $y \in P$ such that $m < y$. Thus, C has a strict upper bound y and we have the following fact:

(iii) Every chain has a strict upper bound.

Define

$$B : \mathcal{C} \to \mathcal{P}(P) \quad \text{by} \quad B(C) = \{b \in P \mid b \text{ is a strict upper bound of } C\}.$$

By fact (iii), every $B(C)$ is nonempty. Let $\mathcal{B} = \operatorname{Im} B$, which is nonempty since $\varnothing \in \mathcal{C}$ and $B(\varnothing) \in \mathcal{B}$. We appeal to the axiom of choice (ZFC9) to invoke a choice function

$$f : \mathcal{B} \to \bigcup \mathcal{B} \quad \text{where} \quad f(B(C)) \in B(C),$$

i.e. $f(B(C))$ is a strict upper bound of C. Change codomains and compose functions to get

$$g = f|^P \circ B|^{\mathcal{B}} : \mathcal{C} \to P.$$

Define a chain C to be **nifty** if the following properties hold:

(N1) $g(\varnothing) \in C$,
(N2) $g(\varnothing) \leq x$ for all $x \in C$,
(N3) $g(C_x) = x$ for all $x \in C$, and
(N4) if X is a strict subchain of C such that $X_x = C_x$ for all $x \in X$, then $g(X) \in C$ and $X = C_{g(X)}$.

Let $\mathcal{N}$ be the collection of all nifty chains and let $N = \bigcup \mathcal{N}$. We go through and prove the following claims.

Claim 1. *For all $C, D \in \mathcal{N}$, either $C \subseteq D$ or $D \subseteq C$, and $C_x = D_x$ for all $x \in C \cap D$.*

Proof. Given $C, D \in \mathcal{N}$, take

$$X = \{x \in C \cap D \mid C_x = D_x\}.$$

Since $X \subseteq C \cap D$, we have that

$$X \subseteq C, \quad X \subseteq D, \quad \text{and} \quad X \text{ is a chain.}$$

Suppose X is a proper subset of C and D. For any $x \in X$, we find $X_x = C_x$ due to the following reasoning:

- $(\subseteq)$. If $y \in X_x$, then $y < x$ and $y \in X \subset C$, so clearly $y \in C_x$.
- $(\supseteq)$. If $y \in C_x$, then $y < x$ and $y \in C$. Now given $C_x = D_x$, it follows that $y \in D$, and a further restriction of these sets yields $C_y = D_y$, so $y \in X$ by definition of X. Given that $y < x$, we have $y \in X_x$.

Thus, X satisfies the hypothesis of (N4), and because C is nifty, we have $g(X) \in C$ and $X = C_{g(X)}$. By the exact same reasoning, $g(X) \in D$ and $X = D_{g(X)}$.

Now $g(X) \in C \cap D$ and $C_{g(X)} = D_{g(X)}$, so then $g(X) \in X$ by definition of X. However, $g(X)$ is a strict upper bound of X, so it cannot be contained in X. This is a contradiction, so the supposition that X is a proper subset of both C and D is false. However, X is a subset of C and D, so $X = C$ or $X = D$. Without loss of generality, assume $X = C$ holds. Given

$$C = X \subseteq C \cap D \subseteq C,$$

it follows that $C = C \cap D$, which is equivalent to $C \subseteq D$. This proves the first part of Claim 1. The fact that $X = C$ proves the second part of Claim 1. □

Claim 2. *N is a chain.*

Proof. The irreflexivity, asymmetry, and transitivity properties of $<$ are trivially passed down from P to N. It only remains to prove the strict comparability property. Let $x, y \in N$. Then $x \in C$ and $y \in D$ for some $C, D \in \mathcal{N}$. By Claim 1, one of the two sets C, D is a subset of the other, say $C \subseteq D$. Then $x, y \in D$ and by the fact that D is a chain it follows that $x < y$, $x = y$, or $x > y$. Thus, any two elements of N are comparable, and therefore N is a chain. This proves Claim 2. □

Claim 3. *For every $C \in \mathcal{N}$, we have $C_x = N_x$ for all $x \in C$.*

Proof. Let $C \in \mathcal{N}$ and $x \in C$. Then $C \subseteq N$, and by Claim 2, actually C is a subchain of N, so $C_x \subseteq N_x$. It remains to show the opposite inclusion $C_x \supseteq N_x$.

Let $y \in N_x$, so $y < x$ and there exists $D \in \mathcal{N}$ such that $y \in D$. By Claim 1, one of the two sets C, D must be a subset of the other. If $D \subseteq C$, then $y \in C$ and with $y < x$ we have $y \in C_x$. If $C \subseteq D$, then $x \in D$ and by Claim 1 again, we have $C_x = D_x$ and so $y \in C_x$. Thus, $N_x \subseteq C_x$, and so $C_x = N_x$, as desired. This proves Claim 3. □

Claim 4. *N is a nifty chain.*

Proof. By Claim 2, N is a chain. We will show that N satisfies the properties (N1)–(N4). Property (N1) holds, because $\{g(\varnothing)\} \in \mathcal{N}$. Property (N2) holds, because (N2) holds for every $C \in \mathcal{N}$.

Now we show (N3) applies to N. Given any $x \in N$, we know $x \in C$ for some $C \in \mathcal{N}$. By Claim 3, $C_x = N_x$, and since (N3) holds for C, we have

$$g(N_x) = g(C_x) = x.$$

This proves that property (N3) holds.

Lastly, we show (N4) applies to N. Let $X \subset N$ such that $X_x = N_x$ for all $x \in X$. Choose any $y \in N \setminus X$. Then $y \in D$ for some $D \in \mathcal{N}$. If $y < x$ for some $x \in X$, then $y \in N_x = X_x$ and $y \in X$, which is a contradiction. Thus, $y \geq x$ for all $x \in X$, i.e. y is an upper bound of X. Given that $y \notin X$, we know $y \neq x$ for all $x \in X$, so y is a strict upper bound of X.

Since $D_y = N_y$ (by Claim 3), $X \subset N$, and y is a strict upper bound of X, we have

$$X \subseteq N_y = D_y \subset D.$$

Let $x \in X$. Since $X_x = N_x$ by assumption and $D_x = N_x$ by Claim 3, it follows that $X_x = D_x$. Thus, $X \subset D$ and for all $x \in X$ we have $X_x = D_x$. Since (N4) holds for D, we have $g(X) \in D$ and $X = D_{g(X)}$. Recalling $D \subseteq N$ (by definition of N) and $D_{g(X)} = N_{g(X)}$ (by Claim 3), we conclude $g(X) \in N$ and $X = N_{g(X)}$. This proves that property (N4) holds and completes the proof of Claim 4. $\square$

Now that we've proven that N is a nifty chain, take the strict upper bound $g(N)$ of N and form the set $N \cup \{g(N)\}$.

Claim 5. *$N \cup \{g(N)\}$ is a nifty chain.*

Proof. Since N is a chain and $g(N)$ is a strict upper bound of N, it follows that $N \cup \{g(N)\}$ is a chain. We will show that $N \cup \{g(N)\}$ satisfies the properties (N1)–(N4). Property (N1) holds, because $g(\varnothing) \in N \subseteq N \cup \{g(N)\}$. Property (N2) holds, because (N2) holds for N, and $g(\varnothing) \leq g(N)$ since $g(N)$ is a strict upper bound of N.

Now we show (N3) applies to $N \cup \{g(N)\}$. Let $x \in N \cup \{g(N)\}$. If $x \in N$, then $(N \cup \{g(N)\})_x = N_x$, because $g(N)$ is a strict upper bound of N. Since (N3) holds for N, we have

$$g((N \cup \{g(N)\})_x) = g(N_x) = x.$$

If $x = g(N)$, then $(N \cup \{g(N)\})_x = N$, so

$$g((N \cup \{g(N)\})_x) = g(N) = x.$$

This proves that property (N3) holds.

Lastly, we show (N4) applies to $N \cup \{g(N)\}$. Let X be a strict subchain of $N \cup \{g(N)\}$ such that $X_x = (N \cup \{g(N)\})_x$ for all $x \in X$. If $g(N) \in X$, then the hypothesis $X_x = (N \cup \{g(N)\})_x$ applies with $x = g(N)$. Thus,

$$X_{g(N)} = (N \cup \{g(N)\})_{g(N)} = N.$$

Hence $N \subseteq X$. Since $g(N) \in X$, it follows that $N \cup \{g(N)\} \subseteq X$. But $X \subseteq N \cup \{g(N)\}$, so $X = N \cup \{g(N)\}$, contradicting the fact that X is a strict subchain. Thus, $g(N) \notin X$, so $X \subseteq N$.

If $X = N$, then

$$g(X) = g(N) \in N \cup \{g(N)\} \quad \text{and} \quad X = N = (N \cup \{g(N)\})_{g(N)}.$$

Thus, property (N4) holds in this case. Now suppose instead that $X \subset N$. For every $x \in X$, we have

$$(N \cup \{g(N)\})_x = N_x,$$

because $g(N)$ is a strict upper bound of N. Hence $X_x = N_x$ for all $x \in X$. Since (N4) holds for N, we have $g(X) \in N$ and $X = N_{g(X)}$. Note that since $g(X) \in N$ and $g(N)$ is a strict upper bound of N, we also have $N_{g(X)} = (N \cup \{g(N)\})_{g(X)}$. Therefore,

$$g(X) \in N \cup \{g(N)\} \quad \text{and} \quad X = (N \cup \{g(N)\})_{g(X)}.$$

This proves that property (N4) holds. This completes the proof of Claim 5. $\square$

By Claim 5, $N \cup \{g(N)\}$ is a nifty chain, so $N \cup \{g(N)\} \in \mathcal{N}$. Since $N = \bigcup \mathcal{N}$, we have $N \cup \{g(N)\} \subseteq N$, and thus $g(N) \in N$. However, $g(N)$ is a strict upper bound of N, so $g(N) \notin N$. This is a contradiction, so our initial supposition that facts (i) and (ii) can hold at the same time is false, which is what we set out to prove. $\blacksquare$

Appendices